

Learning Robust Aligned Representations Across Multiple Visual Modalities in Human Action Recognition

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ABSTRACT

This thesis explores a novel multi-modal co-learning framework for human action recognition, aiming to learn robust representations that can generalize across domains and reduce reliance on annotated data. The proposed approach, named Cross-Modal Video Representation Alignment (CMVRA), aligns different visual modalities, including RGB, infrared (IR), depth, and skeleton data, using contrastive learning techniques. Experiments conducted on the NTU and Drive&Act datasets demonstrate the effectiveness of the framework in capturing generalizable features. Key findings include the investigation of various alignment strategies, and identifying the OMNIVORE model as a versatile option for adapting to new modalities. Fully integrated alignment shows promise in capturing complementary information from multiple modalities when using a base model with rich representations. The study explores the potential of leveraging vision-language models for artificial label generation and introduces the concept of self-alignment, which aligns representations within the same modality using different pre-trained models. Self-alignment experiments reveal insights into the properties of embedding spaces and the role of contrastive learning objectives. The combination of artificial label generation and self-alignment demonstrates the potential of leveraging advanced language models in action recognition tasks. Fine-tuning encoders using captions generated by the LLaVA model maintains or enhances performance across various evaluation tasks compared to using base weights. Zero-shot evaluation on the Drive&Act dataset showcases the generalizability of the learned representations to novel action classes. The thesis contributes valuable insights into multi-modal co-learning for action recognition, highlighting the effectiveness of different alignment strategies and the potential of self-alignment in learning robust representations.

Keywords: Human Action Recognition - Multi-modal Co-Learning - Contrastive Learning - Visual Modality Alignment - Vision-Language Models - Self-Alignment

DEUTSCHE ZUSAMMENFASSUNG

Diese Masterarbeit stellt das neuartige Multi-Modal Co-Learning Framework "Cross-Modal Video Representation Alignment" (CMVRA) für die Erkennung menschlicher Aktivitäten in Videos vor. Ziel ist es, robuste Repräsentationen zu lernen, die auf verschiedene Domänen generalisierbar sind und die Abhängigkeit von annotierten Daten reduzieren. CMVRA baut auf einer Video-Erweiterung des CLIP-Modells auf und richtet die Repräsentationen verschiedener visueller Modalitäten (RGB, Infrarot, Tiefenbilder, Skelett) anhand von Text aus. Durch Contrastive Learning werden die Embeddings in einen gemeinsamen Vektorraum überführt. Experimente auf den Datensätzen NTU und Drive&Act zeigen die Effektivität des Frameworks bei der Erfassung generalisierbarer Merkmale. Die Verwendung modalitätsspezifischer, vortrainierter Encoder wie OMNIVORE führt zu schnellerer Konvergenz und verbesserter Leistung. Ein wesentlicher Beitrag dieser Arbeit ist die Einführung des Self-Alignment-Konzepts, bei dem Repräsentationen innerhalb derselben Modalität unter Verwendung verschiedener vortrainierter Modelle aufeinander abgestimmt werden. Dieser Ansatz ermöglicht es, komplementäre Stärken zu nutzen und robustere Repräsentationen ohne traditionelle Labels zu lernen. Die Integration von Vision-Language-Modellen wie LLaVA zur Erzeugung künstlicher Labels zeigt vielversprechende Ergebnisse bei der Reduzierung der Abhängigkeit von manuellen Annotationen. CMVRA nutzt das RGB-Modell als Anker, um die gelernte Verbindung zwischen RGB-Daten und Textbeschreibungen auf andere visuelle Modalitäten zu übertragen. Die Experimente zeigen, dass dieser Ansatz vielversprechende Ergebnisse erzielt, obwohl die Retrieval-Leistung noch hinter der Linearen Evaluation (LEP) zurückbleibt. Fusionsexperimente auf dem NTU-Datensatz zeigen die Vorteile der Integration von Informationen aus mehreren Modalitäten. Ein einfacher attention basierter Fusionskopf führt zu signifikanten Leistungssteigerungen und erhöhter Robustheit gegenüber fehlenden Modalitäten. Auf dem Drive&Act-Datensatz bleibt die Fusionsleistung jedoch geringer, was weiterer Untersuchung bedarf. Insgesamt stellt CMVRA eine vielversprechende Richtung für Multi-Modal Co-Learning in der Aktivitätserkennung dar. Zukünftige Forschung sollte sich auf die Optimierung des Contrastive Learning Frameworks, die Untersuchung von Einflussfaktoren auf die Fusionsleistung und die Skalierung des Frameworks auf weitere Datensätze. Außerdem sollte der Lernprozess mit durch Vision-Language-Modelle erzeugten Annotationen weiter verfolgt werden, da dieser bereits in unserem einfachen Anwendungsbeispiel vielversprechende Ergebnisse liefern konnte.

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I INTRODUCTION

The field of action recognition has witnessed significant advancements in recent years, driven by the growing demand for intelligent systems capable of interpreting and responding to human actions [SHU22]. Despite these recent advancements, there is a pressing need for models that can learn from diverse modalities and generalize to real-world scenarios with limited resources. The emergence of multi-modal approaches, exemplified by models like CLIP [RKH⁺21] and ImageBind [GENL⁺23], has opened up new possibilities for action recognition. By integrating and processing information from multiple modalities, these models have demonstrated impressive performance in tasks such as zero-shot classification, highlighting the potential of leveraging diverse data sources to enhance action recognition systems. Furthermore, the advent of powerful vision-language models, such as the Large Language and Vision Assistant (LLaVA) [LLWL24, LLLL23], presents an opportunity to bridge the gap between natural language and visual understanding. These models can generate rich, contextual descriptions for visual data, potentially reducing the reliance on manual annotation and facilitating the development of more flexible and adaptable action recognition models. This research focuses on advancing the state-of-the-art in self-supervised action recognition by exploring novel approaches to multi-modal learning and leveraging the capabilities of vision-language models. The main objectives of this work are to develop effective alignment strategies for integrating multiple visual modalities, investigate the practical applications and benefits of the proposed techniques, and explore methods for reducing the dependence on strict labels by leveraging language. The study utilizes two datasets: the larger, high-quality NTU RGB+D dataset [SLNW16] and the more challenging unbalanced Drive&Act (DAA) dataset [MRH⁺19]. By focusing on these datasets, the research aims to develop methodologies that can effectively transfer knowledge from larger, more balanced datasets to improve performance on limited, unbalanced data in a slightly different domain. The main contributions of this work are threefold:

- **Unified multi-modal Embedding Framework:** Created a novel framework that integrates sequential RGB, depth, infrared, skeleton, and text data into a unified embedding for human action recognition.

- **Advancements in Alignment Techniques:** Compared different alignment methods, improving model performance by leveraging complementary information from different modalities. Demonstrated superior performance of fully integrated alignment over pairwise alignment. Achieved faster encoder alignment and improved downstream performance using modality-specific pre-trained models, particularly OMNIVORE for new modalities.
- **Integration of Textual and Visual Data:** Showed how vision-language models can be leveraged for fine-tuning RGB-text alignments, thereby closing the gap between action recognition and vision-language models and enabling a promising angle for future label-free classification across visual modalities.

By pushing the boundaries of self-supervised learning and multi-modal integration, this work aims to contribute to the development of action recognition systems that are more efficient, adaptable, and applicable to a wide range of real-world scenarios.

1.1 PROBLEM FORMULATION

Existing approaches to multi-modal learning, such as ImageBind [GENL⁺23] and the original CLIP [RKH⁺21], have demonstrated impressive performance in aligning and integrating information from multiple modalities. However, these approaches primarily focus on static image data and cannot effectively model temporal dynamics in video sequences. While video extensions of CLIP, such as CLIP-ViP [XSL⁺22], have been developed to address this limitation, they are typically trained on large-scale datasets with millions of text-video pairs and require substantial computational resources, such as batch sizes exceeding 1024. This research explores extending the CLIP framework to integrate temporal data from multiple visual modalities (RGB, depth, skeleton, and infrared) for improved action recognition. Specifically, it introduces a novel approach called "Cross-Modal Video Representation Alignment" (CMVRA), which aims to align the representations learned from different video modalities to capture complementary information and enhance the understanding of human actions in video data. The proposed CMVRA method builds upon video extensions of CLIP to effectively process and align these multi-modal temporal sequences. One of the key challenges that this research wants to address is the limited availability of large-scale, annotated multi-modal video datasets. The study focuses on two smaller datasets, NTU RGB+D and Drive&Act, which present unique challenges in terms of data size and diversity. The first research question aims to determine whether the approaches used in CLIP and its video extensions can be effectively applied to these smaller datasets and how different parameter setups

can be optimized to improve performance. Another important aspect of this research is the exploration of leveraging the text encoder component of CLIP-based models for classification tasks using class labels as text input. By investigating the effectiveness of the text encoder in capturing meaningful semantic information, this study aims to assess the quality of the combined embedding space for classification without modality-specific end-to-end trained classifiers. If the text encoder demonstrates strong performance in this context, it can serve as a bridge to integrate vision-language models into the action recognition pipeline. Building upon this, the research explores the potential of using vision-language models, such as LLaVA [LLLL23, LLWL24], to generate informative textual descriptions for video frames without relying on manual annotations. This opens up new possibilities for label-free classification across visual modalities, significantly reducing the reliance on expensive and time-consuming annotation processes. The successful integration of vision-language models in this context could lead to more efficient and scalable approaches to multi-modal action recognition.

1.2 RESEARCH OBJECTIVES

The primary goal of this research is to investigate the effectiveness and optimal application of self-supervised learning techniques, particularly Visual Modality Alignment, in enhancing the performance and generalizability of action recognition models for video data. The research objectives are categorized into three main parts: (1) effective alignment strategies, (2) practical applications and benefits, and (3) leveraging language to reduce dependence on strict labels.

EFFECTIVE ALIGNMENT STRATEGIES

1. Evaluate the impact of pairwise versus fully integrated alignment strategies on action recognition models' performance and efficiency. This study aims to compare individual RGB-modality alignment against joint alignment of all modalities, while also assessing the potential for inter-modal learning.
2. Investigate the efficiency gains and performance improvements achieved by leveraging modality-specific pre-trained encoders, such as Omnivore, DINO v2, TAHAR, and MAE, compared to training from scratch using the CLIP-ViP model for all modalities. This objective seeks to identify the most effective pre-trained models for each modality and assess their impact on convergence speed and overall performance.
3. Assess the effectiveness of one-way adaptability, where the RGB encoder weights are frozen during alignment, compared to mutual adaptability, where all encoder weights

are trainable. This objective aims to validate whether binding to a strong RGB encoder outperforms free alignment in terms of performance and computational efficiency.

4. Explore the potential of self-alignment to enhance general, strong embeddings without relying on labels. This objective aims to investigate whether aligning representations within the same modality (RGB) using different pre-trained models, such as CLIP-ViP and OMNIVORE, can lead to improved retrieval performance and more discriminative embedding spaces.

PRACTICAL APPLICATIONS AND BENEFITS

1. Evaluate the generalizability of the aligned multi-modal embeddings across datasets (NTU and DAA) and viewpoints (frontal, side, and rear infrared views in DAA). This objective seeks to determine the transferability and robustness of the learned representations in different action recognition scenarios.
2. Compare the performance of the CLIP-ViP language encoder for retrieval tasks against a trained linear evaluation protocol (LEP). This objective aims to assess the practical advantages and potential performance gaps between using the language encoder directly for retrieval and employing a supervised LEP.
3. Investigate the fusion performance and robustness of the aligned encoders in scenarios with missing modalities. This objective seeks to determine whether the encoders capture complementary features during alignment and how well an attention-based fusion mechanism can compensate for missing data.
4. Conduct a zero-shot evaluation on the Drive&Act dataset to assess the generalizability of the learned representations to novel action classes. This objective aims to demonstrate the effectiveness of the proposed approach in handling unseen actions and its potential for open-set recognition scenarios.

LEVERAGING LANGUAGE AND REDUCING DEPENDENCE ON STRICT LABELS

1. Explore the impact of using varied textual descriptions, generated by gpt-4-1106-preview [Ope24] on the performance and alignability of the RGB and IR encoders. This objective aims to assess whether increased variety in descriptions leads to more generalizable representations and improved alignment.
2. Evaluate the effectiveness of using captions generated by the LLaVA model as a substitute for traditional labels in the CLIP-ViP system. This objective seeks to determine

whether synthetic labels can enhance the performance of multi-modal action recognition systems compared to using actual dataset labels.

1.3 THESIS OUTLINE

The core of this thesis is structured into eight main chapters, each building upon the previous to develop a comprehensive understanding of contrastive learning, its application to multi-modal data, and the benefits derived from these methods. Chapter 2 presents a review of existing literature on contrastive learning, multi-modal learning, and action recognition. This chapter discusses key methodologies, notable advancements, and the gaps that this research aims to address. Chapter 3 provides the background information and theoretical underpinnings of the proposed alignment framework for action recognition. It covers the CLIP-ViP model, the description of pre-trained models used in the experiments, and an introduction to the Large Language and Vision Assistant (LLaVA). Chapter 4 details the methods used in the research, including contrastive learning, self-alignment techniques, and the dataset construction process. It describes the multi-modal dataset, CLIP-ViP RGB-Text dataset, and the generation of an RGB-Text dataset from RGB videos. Chapter 5 presents the experimental setup, outlining the overall setup descriptions, fine-tuning of CLIP-ViP, alignment phase, and evaluation phase. It also describes the experiments conducted to improve alignment strategies, such as one-way vs. mutual adaptability, pairwise vs. fully integrated alignment, leveraging modality-specific pre-trained encoders, and self-alignment. Furthermore, it explores the practical benefits and applications of the proposed alignment methods, including cross-dataset and cross-view generalizability, retrieval performance, fusion performance, and robustness to missing modalities. Lastly, it explores leveraging language for alignment, examining the impact of varied textual descriptions and the use of vision-language models for label generation. Chapter 6 presents the results of the experiments conducted in Chapter 5. Chapter 7 discusses the limitations of the current study, identifying areas where further research and improvements are needed. It critically assesses the methodologies employed, the datasets used, and the generalizability of the findings. Chapter 8 concludes the thesis by summarizing the key findings and contributions of the research. It reflects on the potential impact of the work in the field of multi-modal learning and action recognition and highlights the importance of the advancements made in self-supervised alignment techniques. Chapter 9 proposes future research directions. It explores potential avenues for extending and enhancing the proposed alignment techniques, such as investigating alternative loss functions, incorporating temporal information during captioning, and applying the methods to a broader range of datasets.

2 LITERATURE REVIEW

2.1 DEEP LEARNING

Deep learning, a subfield of machine learning, has transformed various domains, including computer vision and action recognition [VDDP18]. At its core, deep learning involves training artificial neural networks with multiple layers to learn hierarchical representations from data [GBC16]. These networks are composed of interconnected nodes, or neurons, organized into layers. Each layer learns to detect specific features or patterns from the input data, with deeper layers capturing increasingly complex and abstract representations [LBH15]. The learning process in deep neural networks is guided by the principle of gradient-based optimization. During training, the network is presented with a large dataset of labeled examples, such as videos annotated with action labels. The network's parameters, or weights, are initially set to random values. As the training data is fed through the network, the output predictions are compared to the ground truth labels, and an error or loss function is computed. The goal is to minimize this loss function by adjusting the network's weights in a direction that reduces the error. This is typically achieved through backpropagation, an algorithm that propagates the error gradient backwards through the network, allowing each layer to update its weights based on its contribution to the final prediction [RHW86]. Over many iterations of this training process, the network gradually learns to recognize relevant patterns and features at different levels of abstraction [BCV13]. In the context of action recognition, the lower layers of the network may learn to detect basic visual features like edges, corners, and textures. Middle layers may learn to recognize more complex patterns like object parts or simple motions. Higher layers can then combine these lower-level features to identify more abstract concepts like human poses, interactions, and ultimately, complete actions [HHP17]. This hierarchical learning capability is particularly well-suited for action recognition tasks, as human actions often involve a complex interplay of visual cues across both spatial and temporal dimensions. By automatically learning multi-scale features from raw video data, deep neural networks can effectively capture the discriminative patterns that distinguish different actions, without relying on handcrafted features or explicit motion modeling [CZ17].

2.2 HUMAN ACTION RECOGNITION

Human action recognition, a key task in computer vision, aims to automatically identify and classify human actions from video data [PKC⁺22]. This task is particularly challenging due to the complex spatiotemporal nature of human actions, as well as variations in appearance, viewpoint, and environmental conditions. Deep learning techniques have revolutionized the field of action recognition, enabling significant progress in recent years [HHP17]. Early works in action recognition relied on handcrafted features like space-time volumes [GBS⁺07], sparse spatio-temporal features [Lap05], and trajectories [WS13]. While these approaches achieved reasonable performance, they often struggled to capture the high-level semantics and long-term dependencies inherent in human actions. The advent of deep learning has provided powerful tools to learn hierarchical representations directly from raw video data, leading to substantial improvements in action recognition accuracy [CZ17]. Deep learning architectures for action recognition can be broadly categorized based on the input data modalities they process, such as RGB videos, skeleton sequences, depth maps, and infrared imagery. Here, we highlight key milestones for each modality type, as discussed in a comprehensive survey by Sun et al. [SKR⁺22].

RGB-BASED METHODS

RGB videos provide rich appearance information for action recognition but are sensitive to variations in viewpoint, background, and illumination. Many deep-learning approaches have been proposed to learn discriminative features from RGB data. Two-stream CNNs [SZ14, BFGV17] employ separate convolutional networks to process spatial (frames) and temporal (optical flow) information, which are then fused for final prediction. RNNs, such as LSTMs [DAHG⁺15], have been used to model long-term temporal dependencies in video sequences. 3D CNNs [TBF⁺15] extend 2D convolutions to the temporal dimension, enabling joint learning of spatiotemporal features. Recently, there has been a growing interest in transformer-based models [GCDZ19, BWT21] primarily because they effectively capture long-range relationships using self-attention techniques.

SKELETON-BASED METHODS

Skeleton data, obtained from motion capture systems or pose estimation algorithms, provides a compact and efficient representation of human actions. Skeleton-based methods are generally robust to variations in appearance and viewpoint. RNNs [DWW15, LWH⁺17] have been widely used to model the temporal dynamics of skeleton sequences. CNNs have been applied to skeleton data by representing it as a sequence of joint coordinate images

[SKR17] or graphs [LAM19]. Graph Convolutional Networks (GCNs) [YXL18, CHC⁺22] have emerged as a powerful tool to capture the structural dependencies between joints in the skeleton. Transformer-based models [SLQ⁺21, ZWL⁺21] have also been explored for modeling the spatial and temporal correlations in skeleton data.

DEPTH-BASED METHODS

Depth maps offer information about the 3D structure of a scene, which can be useful for action recognition, especially in environments with variable lighting conditions. Deep learning methods for depth-based action recognition include multi-stream architectures [XCW⁺19, SLF⁺22] that combine depth and RGB cues, as well as approaches that process depth data directly, such as dynamic depth images [WWG⁺17].

INFRARED-BASED METHODS

Infrared imagery enables action recognition in low-light or nighttime conditions, where RGB data may be less informative. Deep learning approaches for infrared-based action recognition include two-stream CNNs [ASG18] and 3D CNNs [MBA⁺19] that process infrared frames and optical flow.

MULTI-MODALITY METHODS

While single-modality methods have achieved impressive results, there is a growing trend towards leveraging multiple modalities to improve action recognition performance [SKR⁺22]. Multi-modality methods aim to exploit the complementary information provided by different data sources. Fusion strategies combine features [RB17] or predictions [RZG⁺21] from multiple modalities, often leading to improved accuracy and robustness compared to single-modality approaches [SKR⁺22]. Co-learning techniques [SLLG20, GMM19, LWLL21] transfer knowledge between modalities during training, enabling one modality to benefit from the learning of another modality without requiring multi-modal inputs at test time. Despite significant progress, most current action recognition methods rely on supervised learning with large-scale labeled datasets. Cross-dataset evaluation and unsupervised learning from multi-modal data remain challenging and relatively unexplored [SKR⁺22]. This provides an opportunity to leverage advanced representation learning techniques, such as CLIP [RKH⁺21] and its predecessors, to develop more robust and generalizable action recognition models that can learn from diverse, unlabeled data sources.

2.3 MULTI-MODAL CO-LEARNING

Co-learning is a strategy in machine learning that involves transferring knowledge between different modalities during training. This method enhances each modality’s representations without the requirement for multi-modal inference at the test phase. In contrast with fusion techniques that jointly use multiple modalities during testing [SLLG20, GMM19, LWLL21], co-learning has its roots in natural language processing. An early model illustrating this is the model proposed in ”Zero-Shot Learning Through Cross-Modal Transfer” by Socher et al. [SGMN13], where the methodology employs unsupervised text corpora to enrich the visual recognition tasks in a zero-shot learning framework. By mapping images into a semantic space of words, which is grounded in visual modality through the alignment with unsupervised text data, the model can recognize objects in images even without prior training data for those objects. This form of cross-modal knowledge transfer facilitates the recognition of both seen and unseen classes of objects, exemplifying the essence of co-learning where knowledge from the text domain is utilized to enhance the learning in the visual domain, without necessitating multi-modal inference during the testing phase. In the field of computer vision, the CLIP model [RKH⁺21] has emerged as a significant contributor to the domains of zero-shot transfer and multi-modal co-learning. The concept of zero-data learning, which CLIP builds upon, has been explored for over a decade [LEB08, LNH09, LBSF⁺15]. An essential advancement in this area was the realization that natural language could serve as a versatile prediction space, enhancing generalization and transfer capabilities [SGMN13, FCS⁺13]. A particular highlight in this development was the work by Li et al. [LJJM17], which showcased how natural language supervision could enable remarkable zero-shot transfer to ImageNet. Building on this, CLIP, along with other recent studies, revisits the potential of extracting visual representations from textual supervision. This approach is made possible using advanced architectures like Transformers [VSP⁺17] and techniques such as contrastive learning [ZJM⁺22]. A notable extension of the CLIP model is ImageBind [GENL⁺23], which proposes a framework for aligning representations from multiple modalities, including images, audio, depth maps, and text. While ImageBind uses modality-specific encoders, these encoders share the same architecture, enabling the model to learn a unified representation space. However, ImageBind primarily focuses on image-based modalities and has limited capacity for processing video data, relying on just two frames for video understanding. This limitation in ImageBind’s video handling capability underscores the need for effective video extensions of CLIP, which can capture the rich temporal dynamics and multi-modal information present in video data. To address this need, we propose the Cross-Modal Video Representation Alignment (CMVRA) framework, which extends the concepts of multi-modal alignment to the video

domain, specifically targeting human action recognition tasks. By exploring these video extensions and developing the CMVRA framework, we aim to effectively align and integrate multiple video modalities for improved action recognition and understanding.

2.4 VIDEO EXTENSIONS OF CLIP

The success of CLIP in image-text learning has prompted research into its extensions for videos. Notable works include Clip2Video [FXXC21], Clip4Clip [LJZ⁺22], Vita-Clip [WNK⁺23], MA-CLIP [XWH⁺23], CLIP-ViP [XSL⁺22], and models based on frozen CLIP features [LGZ⁺22]. Collectively, these efforts reflect the emphasis on incorporating CLIP’s spatial knowledge, emphasizing efficient temporal modeling, and averting catastrophic forgetting when transitioning to video modalities. This section reviews and contrasts six different approaches. Clip2Video [FXXC21] extends CLIP by adding a temporal difference block to capture motion between frames and a temporal alignment block to align video clips and phrases. It simplifies the video-language learning problem into spatial image-text learning and temporal modeling. Experiments show state-of-the-art results on benchmarks like MSR-VTT, MSVD, and VATEX for text-video retrieval [FXXC21]. Clip4Clip [LJZ⁺22] investigates different similarity calculation approaches based on CLIP including mean pooling, sequential, and tight fusion with Transformer. It finds mean pooling works best for small datasets, while a sequential Transformer is better for large datasets. Additional pretraining on HowTo100M is shown to further improve performance [LJZ⁺22]. Vita-Clip [WNK⁺23] adapts CLIP to video using prompt tuning, adding video summary tokens, frame prompts, and video prompts. It demonstrates strong retrieval results and generalization to long videos. Ensembling with supervised models shows CLIP features are highly complementary [WNK⁺23]. MA-CLIP [XWH⁺23] freezes CLIP and adds lightweight adapters for temporal, multi-modal, and joint adaptation. It also designs a text-guided prototype construction module using attention to enhance video prototypes. This approach achieves state-of-the-art few-shot action recognition results with very few added parameters [XWH⁺23]. Frozen CLIP Models [LGZ⁺22] learn a Transformer decoder on fixed CLIP features to perform spatiotemporal fusion. It introduces a local temporal module using temporal convolutions and cross-frame attention. Despite not fine-tuning CLIP, it shows strong action recognition performance [LGZ⁺22]. AIM [YZX⁺23] extends the capabilities of pre-trained CLIP models by introducing a granular adaptation approach. This technique implements distinct lightweight adapters for spatial, temporal, and combined spatiotemporal processing. By doing so, it enhances the model’s capacity for spatiotemporal modeling while preserving the core CLIP architecture. This granular approach in AIM facilitates a more step-wise adap-

tation of the pre-trained image model to the video action recognition task, potentially allowing for a more controlled and efficient adaptation. Performance on K400 zero shot is comparable between the EVL (initial frozen model) and AIM; however, AIM is even more efficient as it achieves comparable performance with fewer additional parameters [YZX⁺23]. The granular and efficient adaptation strategies showcased by AIM, reflecting the principle of a frozen CLIP, elucidate a promising avenue for advancing Visual Modality Alignment in Unsupervised Action Recognition, potentially facilitating more controlled and resource-efficient model adaptations across varied domains. CLIP-ViP [XSL⁺22] introduces auxiliary in-domain captions and a Video Proxy mechanism, enabling Vision Transformers to handle both images and videos. Through Omnisource Cross-Modal Learning, it facilitates joint representation learning from video-subtitle and frame-caption pairs. CLIP-ViP achieves state-of-the-art results on several benchmarks for text-to-video retrieval, significantly reducing pre-training costs compared to training from scratch [XSL⁺22]. Out of all the presented extensions CLIP-ViP [XSL⁺22] is an ideal candidate to build our model around for several reasons. First, it effectively adapts CLIP’s powerful image-text representations to the video domain with minimal architectural changes, allowing us to leverage pre-trained weights while handling temporal information. Second, its Video Proxy mechanism enables efficient global temporal modeling without significantly increasing computational complexity. The approach maintains the binding to a text encoder which is crucial to allow our framework to achieve classification through retrieval.

2.5 UNIVERSAL ENCODERS FOR MULTIPLE MODALITIES

Recent research focused on the development of universal encoders capable of processing multiple modalities. These encoders aim to learn a shared representation space that effectively captures the information from various data sources, such as images, videos, audio, and 3D data. The prospect of using a single model to handle diverse modalities is particularly appealing, as it can greatly simplify the process of building multi-modal AI systems. OMNIVORE [GSR⁺22] is a notable example of this trend, using a shared transformer architecture to process images, videos, and single-view 3D data. Despite the shared architecture, OMNIVORE maintains a single set of weights that is used across all modalities, allowing for a compact and efficient model. In contrast, ImageBind [GENL⁺23] proposes a framework for aligning representations from multiple modalities, including images, videos, audio, depth maps, and text. While ImageBind also uses a shared architecture for all modalities, it employs modality-specific encoders, each with its own set of weights. PolyViT [LAC⁺21] extends the Vision Transformer (ViT) architecture to handle images, videos, and audio data. Similar to Image-

Bind, PolyViT uses a shared transformer architecture but maintains separate sets of weights for each modality, enabling the model to learn a joint representation space while preserving the unique properties of each modality.

2.6 DRIVE&ACT DATASET

The Drive&Act dataset [MRH⁺19] is a novel multi-modal dataset designed for fine-grained driver behavior recognition within vehicles. The dataset features over 9.6 million frames of video data captured from six synchronized camera views, providing a comprehensive understanding of driver behavior and interactions inside the vehicle cabin. As shown in Figure 2.1, the dataset includes a Microsoft Kinect sensor positioned at the bottom left of the cabin, which captures RGB, infrared, and depth data. Additionally, five near-infrared (NIR) cameras are placed at strategic locations: the steering wheel (Face-view), driver side (Left-top), co-driver side (Right-top), ceiling (Back), and inner mirror (Front-top). These camera views are designed to cover different angles and regions of interest within the vehicle cabin. Furthermore, the dataset provides 3D body and head pose information, which is initially generated from the Inner Mirror view but essentially covers every angle. The inclusion of RGB, infrared, and depth modalities, along with the pose data, enables the development of robust activity recognition models that can perform well under various lighting conditions and provide valuable insights into the driver’s posture, gaze, and attention. A key characteristic of the Drive&Act dataset is the hierarchical annotation scheme, which provides labels at three levels of granularity. At the coarsest level, the dataset includes 12 high-level scenarios and tasks that drivers had to complete, such as eating, working on a laptop, or responding to a sudden takeover request. At the second, fine-grained level, the dataset provides 34 distinct activity classes, such as opening a bottle, reading a magazine, or fastening a seatbelt. Finally, at the lowest level, the dataset includes 372 possible combinations of atomic actions, objects, and locations. This thesis focuses exclusively on the 34 fine-grained activities, which provide a detailed and nuanced representation of driver behavior. These activities cover a wide range of common driver actions, such as using a phone, eating, drinking, and interacting with various objects within the vehicle cabin. One of the most notable aspects of the Drive&Act dataset is the significant class imbalance present in its fine-grained activity classes. The frequency of individual classes varies dramatically, reflecting the diverse nature of in-vehicle activities. As illustrated in Figure 2.2, this imbalance is particularly evident when comparing the least and most represented classes. The activity ‘taking laptop from backpack’ has only 19 samples, making it the rarest class, while ‘sitting still’ dominates the dataset with 2,797 samples. This extreme imbalance reflects the natural distribution of driver behaviors, where some activities

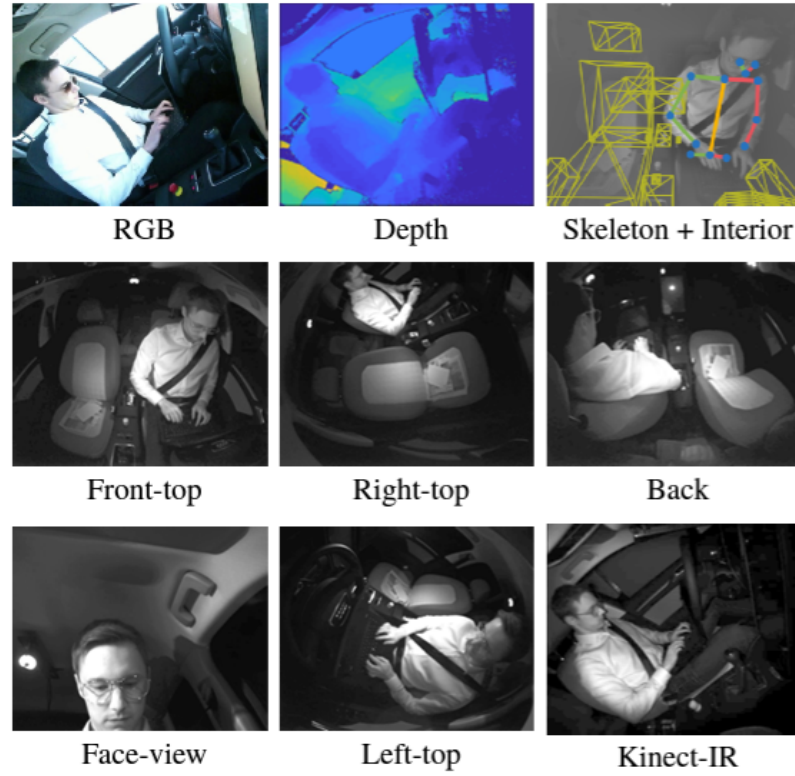


Figure 2.1: Example images of the “working on laptop” activity from different camera views and modalities in the Drive&Act dataset. Reprinted from [MRH⁺19]

are much more common than others, but poses a significant challenge for training effective activity recognition models. Overall, the Drive&Act dataset, with its fine-grained taxonomy of driver behaviors, multi-modal data from various camera angles, and challenging class imbalance, presents a compelling benchmark for advancing the state-of-the-art in-cabin activity recognition.

2.7 APPROACHES ON DRIVE&ACT

The Drive&Act dataset [MRH⁺19] has served as a testbed for exploring distinct approaches in driver activity recognition, particularly focusing on standard classification and extending to scenarios where labeled data is sparse. The latter scenario reflects an effort to bridge the domain gap between controlled, lab-centric models and real-world applications.

Peng et al.’s TransDARC model [PRY⁺22], which uses the Video Swin Transformer architecture [LNC⁺22] for feature extraction, currently holds the state-of-the-art for top-1 accu-

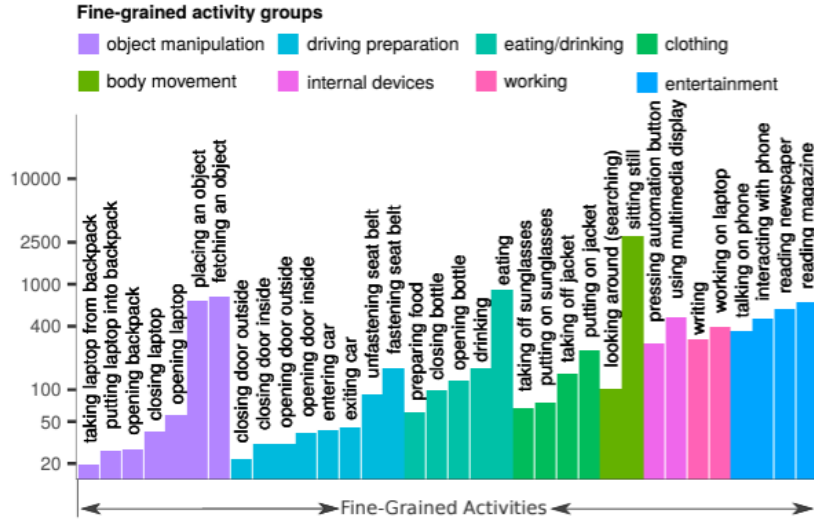


Figure 2.2: Sample frequency of fine-grained activities by class in the Drive&Act dataset. Reprinted from [MRH⁺19].

racy in standard classification. They introduce a novel latent space feature calibration module that augments the training data by interpolating between feature clusters, thereby enhancing generalization. TransDARC significantly outperforms previous models on fine-grained driver activity recognition, the primary benchmark on Drive&Act. Venturing beyond standard classification, Roitberg et al. present Open-Drive&Act, an extension of Drive&Act tailored for open set driver activity recognition [RMHS20]. This framework addresses the challenge of limited labeled data by segregating the 34 activities into 14 "seen" and 2 times 10 "unseen" classes, simulating novel behaviors. They propose a dual-component framework: a novelty detection mechanism coupled with an I3D network for classifying seen behaviors. The novelty detection, enhanced by uncertainty-based approaches like selective voting, significantly outperforms conventional methods, marking a stride towards bridging the domain gap from lab to real-world scenarios.

2.8 NTU RGB+D DATASET

NTU RGB+D 60

The NTU RGB+D 60 dataset [SLNW16] is a comprehensive resource for multi-modal human action recognition research. It includes 56,880 video samples across 60 different action categories, recorded from 40 individuals using Microsoft Kinect v2 sensors. Similar to the Drive&Act dataset [MRH⁺19], NTU RGB+D 60 provides synchronized RGB videos,

depth maps, 3D joint information, and infrared sequences. The dataset defines two evalua-

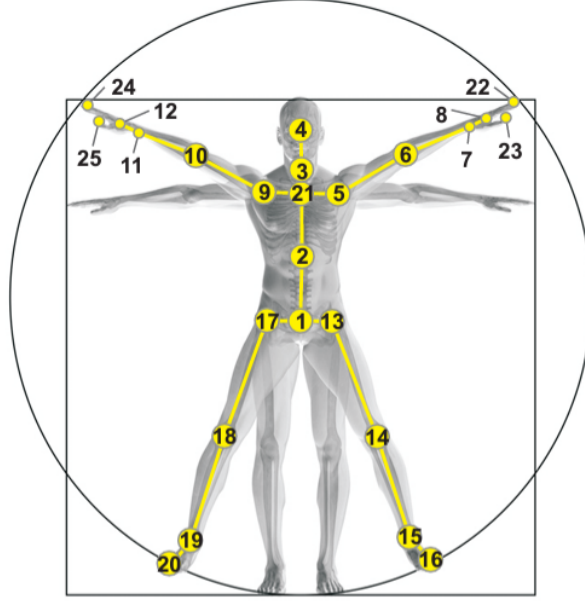


Figure 2.3: The figure illustrates the configuration of 25 body joints used in the NTU RGB+D dataset. These joints include key points such as the base of the spine (1), neck (3), left shoulder (5), right wrist (11), and left ankle (15), among others. Reprinted from [SLNW16].

tion protocols [SLNW16]:

1. Cross-subject: 40 subjects are evenly split into two groups for training and testing, with 20 subjects in each group.
2. Cross-view: Left and right 45° views for testing, frontal and side views for training.

NTU RGB+D 120

Liu et al. [LSP⁺19] extended NTU RGB+D 60 to create NTU RGB+D 120, containing 114,480 samples of 120 action classes, captured from 106 subjects. It includes 82 daily actions, 12 health-related actions, and 26 mutual interactions. NTU RGB+D 120 introduces 32 camera setups varying in height and distance, providing viewpoint diversity for assessing view-invariance. The evaluation protocols are adapted for the larger dataset:

1. Cross-subject: 106 subjects divided into training and testing groups of 53 each.
2. Cross-setup: Even setup IDs for training, odd setup IDs for testing.

The NTU RGB+D dataset, in both its 60 and 120 versions, provides a comprehensive benchmark for human action recognition. Its multi-modal nature, diverse action classes, viewpoint variations, and standard evaluation protocols make it valuable for research on multi-modal alignment.

3 BACKGROUND

3.1 VISION TRANSFORMER

The Vision Transformer (ViT) [DBK⁺20] represents a significant shift in computer vision, applying principles from natural language processing to image analysis. Its development follows the evolution of language models, which progressed from neural probabilistic approaches [BDV00] to sequence-to-sequence learning [SVL14], and ultimately to the Transformer architecture [VSP⁺17]. The key innovation of the Transformer was its reliance on attention mechanisms, changing recurrent structures in favor of parallel processing. This "attention is all you need" philosophy proved remarkably effective for language tasks and inspired researchers to apply similar principles to vision. ViT adapts the Transformer architecture for image classification by treating an image as a sequence of patches. Given an input image $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$, where H , W , and C are the height, width, and number of channels respectively, ViT first reshapes it into a sequence of flattened 2D patches $\mathbf{x}_p \in \mathbb{R}^{N \times (P^2 \cdot C)}$ [DBK⁺20]. Here, (P, P) is the resolution of each patch, typically 16x16 pixels, and $N = HW/P^2$ is the number of patches [DBK⁺20]. These patches are then linearly embedded and combined with position embeddings to retain spatial information [DBK⁺20]:

$$\mathbf{z}_0 = [\mathbf{x}_{\text{class}}; \mathbf{x}_p^1 \mathbf{E}; \mathbf{x}_p^2 \mathbf{E}; \dots; \mathbf{x}_p^N \mathbf{E}] + \mathbf{E}_{\text{pos}} \quad (3.1)$$

where $\mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times D}$ is the patch embedding projection matrix, $\mathbf{x}_{\text{class}}$ is a learnable classification token, and $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{(N+1) \times D}$ are learnable position embeddings [DBK⁺20]. The resulting sequence is processed by a standard Transformer encoder [VSP⁺17], which includes L layers of alternating Multi-head Self-Attention (MSA) and Multi-Layer Perceptron (MLP) blocks:

$$\mathbf{z}'_l = \text{MSA}(\text{LN}(\mathbf{z}_{l-1})) + \mathbf{z}_{l-1} \quad (3.2)$$

$$\mathbf{z}_l = \text{MLP}(\text{LN}(\mathbf{z}'_l)) + \mathbf{z}'_l \quad (3.3)$$

3 Background

where LN denotes Layer Normalization [BKH16]. The MLP is a simple feed-forward neural network typically consisting of two linear transformations with a non-linearity in between [R⁺62, RHW86]. The self-attention mechanism in each MSA block is defined as [VSP⁺17]:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \quad (3.4)$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are query, key, and value matrices, and d_k is the dimension of the key vectors. A notable aspect of ViT is its use of a special "class token", similar to the [CLS] token in BERT [DCLT18]. This token, prepended to the sequence of patch embeddings, aggregates information from the entire image and is used for the final classification decision:

$$\mathbf{y} = \text{LN}(\mathbf{z}_L^0) \quad (3.5)$$

Despite its departure from traditional convolutional architectures, ViT demonstrated competitive performance on image classification benchmarks, particularly when pre-trained on large datasets. Its success paved the way for further explorations in applying Transformer architectures to vision tasks, including the development of multi-modal models like CLIP-ViP [XSL⁺22] and OMNIVORE [GSR⁺22], which we will explore in subsequent sections.

3.2 CLIP-ViP: ADAPTING CLIP FOR VIDEO-LANGUAGE TASKS

CLIP-ViP [XSL⁺22] extends the CLIP model to effectively handle video data while preserving its powerful image-text representations. The key innovation is the **Video Proxy** mechanism, which enables temporal modeling with minimal changes to the original architecture. To adapt CLIP's Vision Transformer (ViT) for video processing, CLIP-ViP introduces learnable video proxy tokens. Given a video with T frames, each divided into N patches, the input sequence becomes [XSL⁺22]:

$$\mathbf{z}_0 = [\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_M, \mathbf{x}_1^1, \dots, \mathbf{x}_N^1, \dots, \mathbf{x}_1^T, \dots, \mathbf{x}_N^T] + \mathbf{E}_{\text{pos}} \quad (3.6)$$

where $\mathbf{p}_i \in \mathbb{R}^D$ are the M video proxy tokens, $\mathbf{x}_n^t \in \mathbb{R}^D$ are the patch embeddings, and $\mathbf{E}_{\text{pos}}$ contains both spatial and temporal position embeddings. The proxy-guided attention mechanism is implemented using an attention mask **MViP** [XSL⁺22]:

$$\mathbf{MViP}(i, j) = \begin{cases} 1 & \text{if } i \leq M \text{ or } j \leq M \text{ or } \lfloor \frac{i-M}{N} \rfloor = \lfloor \frac{j-M}{N} \rfloor \\ 0 & \text{otherwise} \end{cases} \quad (3.7)$$

This mask ensures that video proxies attend to all tokens, while patch tokens only attend within their frame and to the proxies [XSL⁺22]. This design allows the model to capture global temporal information through the proxy tokens while maintaining the spatial structure within each frame. The Video Proxy mechanism adds only $M \times D$ parameters (e.g.,

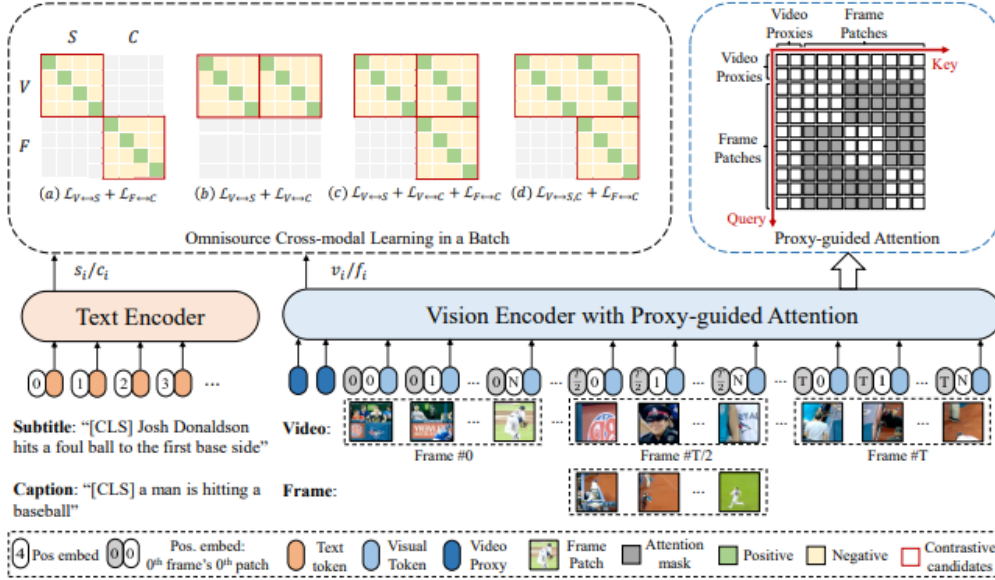


Figure 3.1: CLIP-ViP framework architecture featuring text and vision encoders. The model processes video (V), frame (F), subtitle (S), and caption (C) features. It explores various Omnisource cross-modal learning configurations: (a) $L_{V \leftrightarrow S} + L_{F \leftrightarrow C}$, (b) $L_{V \leftrightarrow S} + L_{V \leftrightarrow C}$, (c) $L_{V \leftrightarrow S} + L_{V \leftrightarrow C} + L_{F \leftrightarrow C}$, and (d) $L_{V \leftrightarrow S, C} + L_{F \leftrightarrow C}$. The vision encoder incorporates Video proxy tokens and a ViP-guided attention mechanism to adapt CLIP for video processing. Reprinted from [XSL⁺22]

3K for $M = 4$, $D = 768$), compared to the 86M parameters of ViT-B/32. This minimal modification allows effective leveraging of CLIP’s pre-trained weights while enabling video processing. CLIP-ViP is an ideal candidate to build our model around for several reasons. First, it effectively adapts CLIP’s powerful image-text representations to the video domain with minimal architectural changes, allowing us to leverage pre-trained weights while handling temporal information. Second, its Video Proxy mechanism enables efficient global temporal modeling without significantly increasing computational complexity. The approach maintains the structure of CLIP’s ViT backbone, ensuring compatibility with existing CLIP models and weights by also keeping the binding to the text encoder. Additionally, the minimal increase in parameters means that duplicating the model for various modalities keeps computational demands manageable. It is worth noting that CLIP-ViP also introduces an Omnisource Cross-modal Learning (OCL) approach, which leverages both original video

subtitles and generated frame captions to enhance video-language alignment. While we do not utilize this aspect in our current work, it presents an interesting direction for future research in multi-modal learning. These characteristics make CLIP-ViP a robust and versatile foundation for developing advanced action recognition systems. Its ability to process video data while leveraging the strong visual representations learned by CLIP provides an excellent starting point for our work.

3.3 DESCRIPTION OF PRE-TRAINED MODELS

For our experiments, we carefully selected several state-of-the-art, publicly available, transformer-based models that are designed to handle specific modalities or demonstrate strong performance in temporal modeling. Each of these models brings unique capabilities that make them well-suited for our alignment experiments:

1. **OMNIVORE**[GSR⁺22] This model is designed to process multiple visual modalities, including images, videos, and single-view 3D data, within a single shared transformer architecture. Its unified feature representation and ability to handle depth data make it a promising candidate for RGB, IR, and depth modalities.
2. **DINOv2**:[ODM⁺23] Trained using self-supervised learning on a large-scale image dataset, DINOv2 has shown remarkable performance across various downstream tasks. Although primarily an image model, its robust feature learning capabilities make it a strong contender for extracting features from individual video frames in RGB and IR modalities.
3. **OmniMAE (Masked Autoencoder)**[GENS⁺23]: Adapted from the concept of masked language modeling in NLP, MAE models learn to reconstruct missing parts of visual inputs. This approach enables learning from large-scale, unlabeled datasets, making it suitable for RGB, IR, and depth modalities, particularly when dealing with limited annotations.
4. **TAHAR (Temporal Adaptive Hierarchical Autoencoder for Skeleton)**[LZM⁺24]: Designed specifically for skeleton data, TAHAR takes into account the hierarchical structure of the human body and the temporal dependencies between frames. Its specialized architecture and training procedure make it an ideal choice for the skeleton modality.

By leveraging these diverse pre-trained models, each with its unique strengths and specializations, we aim to explore their potential in facilitating faster convergence and improved

performance during the alignment phase. The selection of these models also allows us to investigate the effectiveness of modality-specific pre-training in the context of our multi-modal action recognition framework. The following sections provide an in-depth description of each model. The exploration will cover their architectures, training methods, and unique features that contribute to their potential role and impact in the experimental setup.

OMNIVORE

OMNIVORE [GSR⁺22] is a transformer-based architecture designed to process multiple visual modalities, including images, videos, and single-view 3D data, within a single shared model. The key innovation in OMNIVORE is its unified patch embedding mechanism that allows the model to handle diverse input formats while maintaining a shared set of parameters for the transformer layers [GSR⁺22]. The unified patch embedding mechanism is illustrated in figure 3.2. Given an input tensor $\mathbf{X} \in \mathbb{R}^{T \times H \times W \times C}$, where T represents the temporal dimension, H and W are the spatial dimensions, and C is the number of channels, OMNIVORE first splits the input into a sequence of patches [GSR⁺22, DBK⁺20]. Images are divided into non-overlapping spatial patches, videos into spatio-temporal patches, and single-view 3D data is processed by splitting the RGB and depth channels separately [GSR⁺22]. These patches are then projected into a shared embedding space using modality-specific linear layers followed by Layer Normalization (LN) [BKH16]. The RGB patches from images and videos are processed using the same linear layer, while the depth patches are handled by a separate linear layer [GSR⁺22]. The resulting patch embeddings are concatenated along the sequence dimension, forming the input to the transformer layers. The transformer layers in OMNIVORE follow the standard architecture [VSP⁺17], with multi-head self-attention and feed-forward networks. The self-attention mechanism allows the model to capture dependencies between patches across spatial and temporal dimensions [DBK⁺20]. OMNIVORE employs learnable positional embeddings for both spatial and temporal dimensions to incorporate positional information [GSR⁺22]. By using a unified patch embedding mechanism and shared transformer layers, OMNIVORE can learn a joint representation space for multiple visual modalities [GSR⁺22]. During training, OMNIVORE is jointly optimized on standard datasets for each modality, such as ImageNet for image classification, Kinetics for video recognition, and SUN RGB-D for single-view 3D data [GSR⁺22]. The model is trained using a cross-entropy loss for each task, with the losses being summed across all modalities [GSR⁺22]. The unified architecture and joint training strategy of OMNIVORE suggest that it could be a promising initialization candidate for our multi-modal alignment tasks. Its ability to process temporal and depth information within a shared framework has the potential to enable more efficient learning and faster convergence when aligning RGB and depth

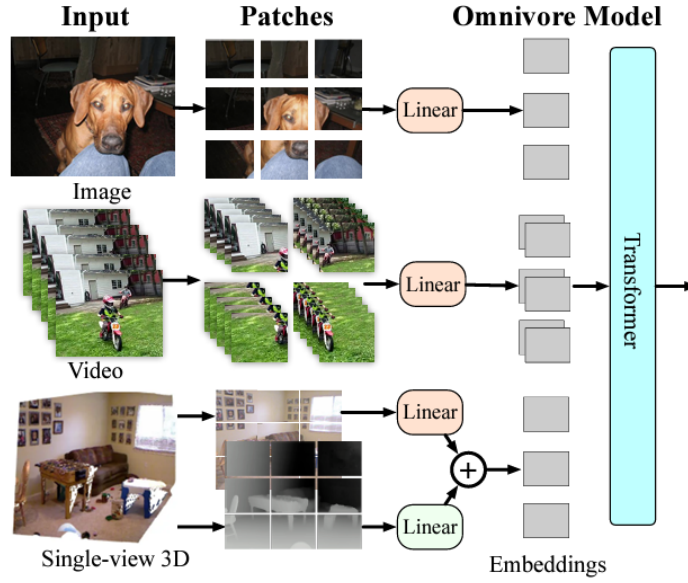


Figure 3.2: Diagram of patch embedding in the OMNIVORE model, illustrating the separate processing of RGB and depth patches for images, videos, and single-view 3D data. The same linear layer is used for RGB patches from both images and videos, while a distinct linear layer processes depth patches, underscoring the model’s specialized approach to handling different data modalities. Reprinted from [GSR⁺22]

modalities. Moreover, the model’s exposure to diverse modalities during training may provide a more robust foundation for adapting to new modalities, such as infrared, compared to models like CLIP-ViP that focus primarily on RGB and video data [XSL⁺22].

DINOv2

DINOv2 is an extension of the original DINO (self-DIstillation with NO labels) [CTM⁺21] which is a self-supervised learning approach designed for training Vision Transformers (ViTs) using unlabeled image data. The core idea behind DINO is to use self-distillation, where a student network learns to match the output of a teacher network, which is an exponential moving average of the student network’s parameters [CTM⁺21]. This self-distillation is inspired by the general idea of knowledge distillation [HVD15]. Building upon the success of DINO, the DINOv2 model, proposed by Oquab et al. [ODM⁺23], introduces several enhancements to further improve the quality and efficiency of self-supervised learning for vision transformers. The model integrates iBOT [ZWW⁺21] with the original DINO [CTM⁺21], while introducing additional techniques to enhance feature quality and training efficiency. These include the centering mechanism from SwAV [CMM⁺20], a regularizer to encourage

uniform feature distribution, and a final high-resolution fine-tuning stage. The pretraining dataset, LVD-142M, is constructed by automatically filtering and balancing a vast collection of uncurated web images, resulting in a diverse and high-quality dataset of 142M images. The DINOv2 models, especially the 1B-parameter vision transformer (ViT-g), achieve remarkable performance across a broad spectrum of downstream tasks at both image and pixel levels, often outperforming the best publicly available self-supervised models and rivaling or even surpassing weakly-supervised models without the need for task-specific fine-tuning. Nonetheless, it is important to note that DINOv2 is only trained on RGB and depth images and, therefore, lacks any kind of temporal modeling. Despite this limitation, DINOv2 could offer a valuable initialization candidate for a multi-modal alignment approach, as its strong performance on image-level tasks suggests that it learns robust and transferable visual features [ODM⁺23].

OMNiMAE

Masked autoencoding, the idea of training models to reconstruct missing parts of the input, originates from the domain of natural language processing (NLP). This approach has been widely used in NLP tasks to enable models to learn contextual representations of words in sentences, such as in the BERT model [DCLT18]. The concept was later adapted for visual data, based on the notion that similar principles can be applied to learning representations from images or videos by predicting missing parts of the visual input [HCX⁺22]. A notable application of masked autoencoding in computer vision is OmniMAE [GENS⁺23], a model that leverages a unified architecture to process both images and videos. Traditional approaches to visual representation learning often relied on separate models for different modalities, each designed to handle the specific characteristics of the input data. In contrast, OmniMAE employs a single ViT backbone to process both images and videos by treating them as sequences of patches. During training, a significant portion of these patches – up to 90% for images and 95% for videos – are randomly masked, and the model is tasked with reconstructing the original visual input. OmniMAE’s training procedure follows an encoder-decoder framework. The encoder processes only the unmasked patches, learning to generate rich, contextual embeddings that capture the spatial and temporal relationships between the visible parts of the input. To maintain the positional information of the masked patches, a learnable “mask token” is introduced during training, acting as a placeholder for the missing information. The decoder, conditioned on the embeddings produced by the encoder, reconstructs the entire visual input, including both the masked and unmasked patches. The training objective is to minimize the reconstruction error between the original input and the decoder’s output, encouraging the model to learn meaningful representations that can gen-

eralize to unseen data [GENS⁺23]. The key general advantage of this approach is that it enables the model to learn from large-scale, unlabeled datasets, as the masking and reconstruction tasks do not rely on explicit annotations [HCX⁺22]. For our alignment experiments, we adapt the MAE concept by training separate models for each modality (RGB, IR, and Depth) and data split. This approach allows each encoder to be specifically tailored to the characteristics of its respective modality and dataset, potentially enabling faster convergence during the alignment phase. By leveraging modality-specific pre-trained encoders, we aim to provide a strong foundation for our multi-modal action recognition pipeline, facilitating the alignment of representations across different modalities. Training and adoption details are reported in Appendix 1.

TAHAR

For the skeleton modality, we employ TAHAR (Temporal Adaptive Hierarchical Autoencoder for Skeleton) [LZM⁺24] instead of OmniMAE. TAHAR is specifically designed for skeleton data, accounting for the hierarchical structure of the human body and temporal dependencies in motion sequences. This specialized approach allows us to leverage a pre-trained model that is better suited to the unique characteristics of skeleton-based representations. The model addresses the challenges posed by skeleton data, such as complex spatial-temporal dependencies and variations in human anatomy and motion, by employing a dual-stream input approach. Both the encoder and the decoder of TAHAR process the 3D skeleton sequence, enabling the model to effectively capture and utilize the intricate spatial and temporal relationships inherent in human movements. The encoder of TAHAR embeds the skeleton sequence into a continuous hidden space and incorporates a learnable CLS token to summarize sequence-level information. This CLS token plays a crucial role in the action recognition phase. The decoder, on the other hand, reconstructs the original skeleton sequence from a masked and noisy version of the input. By training the decoder to reconstruct the original sequence, the model learns to capture robust features and denoise the data, which is particularly useful when dealing with real-world noisy environments [LZM⁺24]. One of the key features of TAHAR is its ability to perform conditioned data generation. This generative capability allows the model to synthesize new, realistic skeleton sequences based on learned conditions. This is particularly beneficial when working with limited or imbalanced datasets, as it can augment the training data and improve the model’s performance. Overall TAHAR and its weights are a promising candidate for a pre-trained skeleton model.

3.4 LARGE LANGUAGE AND VISION ASSISTANT-LLaVA

The development of vision-language models has been driven by the goal of creating AI systems that can effectively understand and reason about the world by integrating visual and linguistic information, mimicking human perception and communication [LLWL24]. Traditional AI approaches often treated vision and language as separate modalities, but in real-world scenarios, they are often intertwined. The ability to jointly reason over both modalities is crucial for many applications, such as image captioning, visual question answering, and multi-modal dialogue systems [LLWL24]. Vision-language models bridge this gap by learning unified representations that capture the semantic relationships between visual and textual inputs, enabling them to perform complex tasks that require reasoning, grounding, and cross-modal alignment [LLWL24]. The success of large language models (LLMs) in natural language processing has also highlighted the potential of scaling up models and training on vast amounts of data to achieve impressive performance across a wide range of tasks, inspiring researchers to explore similar approaches in the multi-modal domain [LLWL24]. LLaVA (Large Language and Vision Assistant) is a recent approach that exemplifies this trend of building large multi-modal models capable of understanding and reasoning about visual and linguistic information [LLWL24, LLLL23]. LLaVA leverages the power of pre-trained vision models and LLMs by aligning their representations through visual instruction tuning [LLWL24, LLLL23]. It uses a pre-trained vision encoder (e.g., CLIP) to extract visual features from images and a pre-trained LLM (e.g. Hermes-Yi-34B) to process and generate natural language. A vision-language connector module is introduced to map the visual features into the language model’s embedding space [LLWL24, LLLL23]. The alignment in LLaVA is performed through a two-stage training process. First, the vision-language connector is pre-trained on a large dataset of image-text pairs to establish a common representational space. Second, the entire model is fine-tuned on a smaller dataset of visual instruction-following tasks to adapt it to specific downstream applications [LLWL24, LLLL23]. This approach allows LLaVA to leverage the strengths of both pre-trained vision models and LLMs.

4 METHODS

This chapter presents the key methodological components of our Cross-Modal Video Representation Alignment (CMVRA) framework. We start by discussing the theoretical foundations of contrastive learning and introduce our novel extension of the InfoNCE loss to handle multiple modalities simultaneously. We then describe the concept of self-alignment, which leverages the complementary strengths of different pre-trained models within a single modality. In addition, we detail our dataset construction process, including the creation of a multi-modal dataset, the construction of a CLIP-ViP RGB-Text dataset for fine-tuning, and the generation of an RGB-Text dataset from RGB videos using the LLaVA model.

4.1 CONTRASTIVE LEARNING

This section outlines the theoretical and technical setup underpinning the experiments detailed later, focusing on contrastive learning methods, particularly the InfoNCE loss, which is pivotal in unsupervised and self-supervised learning paradigms for learning effective representations from unlabeled data. Introduced by van den Oord et al. (2018) in their work on Contrastive Predictive Coding (CPC), InfoNCE loss represents a method for maximizing mutual information between samples in a dataset without requiring explicit labels [OLV18]. Contrastive learning aims to learn representations by distinguishing a target sample from contrasting (noise) samples. The InfoNCE loss, a specific form of contrastive loss, is designed to maximize the mutual information between related samples and minimize information with unrelated samples. Mathematically, it is defined as:

$$\mathcal{L}_{\text{InfoNCE}} = -\mathbb{E} \left[\log \frac{\exp(f(x_t, c_t))}{\sum_{x_j \in \mathcal{X}} \exp(f(x_j, c_t))} \right] \quad (4.1)$$

Here, x_t is the target sample, c_t represents the context or condition under which x_t and other samples x_j are evaluated, and f is a scoring function assessing the compatibility between samples and their context. The set $\mathcal{X}$ includes one positive and several negative samples, typically drawn from a noise distribution. In ImageBind [GENL⁺23], the InfoNCE loss is modified to align embeddings from distinct modalities into a coherent shared embedding space, facil-

itating cross-modal retrieval via cosine similarity. ImageBind introduces a bidirectional version of the loss, ensuring that embeddings from each modality can predict and be predicted by the other. The bidirectional loss formulation, integrating a temperature parameter (τ), is described as follows:

$$\mathcal{L}_{I,M} + \mathcal{L}_{M,I} = - \left[\log \frac{\exp(q_i^\top k_i / \tau)}{\exp(q_i^\top k_i / \tau) + \sum_{j \neq i} \exp(q_i^\top k_j / \tau)} + \log \frac{\exp(k_i^\top q_i / \tau)}{\exp(k_i^\top q_i / \tau) + \sum_{j \neq i} \exp(k_i^\top q_j / \tau)} \right] \quad (4.2)$$

Here, $q_i = f(I_i)$ and $k_i = g(M_i)$ are the normalized embeddings of an image I_i and a modality M_i respectively, encoded by the neural networks f and g [GENL⁺23]. This symmetrical loss formulation ensures a balanced and unbiased learning process across modalities. Building upon this symmetrical InfoNCE loss, we introduce a novel extension to handle more than two modalities in the alignment process. Our method considers all pairwise combinations of modalities and computes the total loss as the average of the individual InfoNCE losses for each pair:

$$\mathcal{L}_{\text{total}} = \frac{1}{M(M-1)/2} \sum_{i=1}^{M-1} \sum_{j=i+1}^M \mathcal{L}_{\text{InfoNCE}}(\mathbf{z}_i, \mathbf{z}_j) \quad (4.3)$$

where M is the total number of modalities, and $\mathbf{z}_i$ and $\mathbf{z}_j$ are the embeddings from modalities i and j , respectively. By optimizing this total loss, our model simultaneously aligns the embeddings from all modalities, enabling it to capture shared semantic information while preserving the unique characteristics of each modality. This novel formulation of the InfoNCE loss for multiple modalities is a key contribution of our work, as it allows for the effective integration of complementary information from various data sources in a unified embedding space. We demonstrate the efficacy of this approach through a series of experiments detailed in Section 5.2.2. To efficiently implement the contrastive learning approach, we use a parent model with specific encoders for each modality (or different views for DAA) M_{iv} . For an experiment, it is required to have at least two distinct modalities (or views). We use the RGB encoder as a strong base for most of the following experiments, as ImageBind demonstrated the effectiveness of using CLIP weights as an anchor for alignment [GENL⁺23]. Additionally, aligning to a frozen video RGB encoder might lead to emergent alignment towards text, similar to ImageBind. We validate the choice of using a frozen RGB encoder versus having both encoders trainable during alignment in section 5.1.

SELF-ALIGNMENT

In addition to cross-modal alignment, we propose the concept of self-alignment, which involves aligning representations obtained from different pre-trained models within a single modality. This approach is motivated by several observations and draws inspiration from the knowledge distillation process [HVD15]. Firstly, different pre-trained models, even when trained on the same modality, can capture diverse and complementary representations. This is due to factors such as differences in architecture, training objectives, and datasets used during pre-training. By aligning these representations, we aim to leverage the unique strengths of each model and obtain a more comprehensive and robust representation of the target modality. Secondly, self-alignment can be seen as a form of knowledge distillation, where the knowledge from a text-aligned model is transferred to another pre-trained model. In this case, the aligned representation can be considered as the student model that learns to capture the relevant and informative features from the text-aligned teacher model, potentially enhancing its performance without requiring additional text annotations. Thirdly, self-alignment has the potential to improve the generalization and transferability of the learned representations. By combining the knowledge from multiple pre-trained models, the aligned representation may be more resilient to variations in data distribution and better equipped to handle downstream tasks that differ from the pre-training domain. Lastly, in scenarios where one encoder is fixed and the other is frozen during the alignment process, the contrastive loss used in self-alignment can lead to a more spread-out embedding space. This is because the off-diagonal values in the cosine similarity matrix, which represent the similarities between non-corresponding samples, are pushed apart by the contrastive loss. As a result, the self-aligned representations are encouraged to be more diverse and discriminative, potentially enhancing their effectiveness in downstream tasks. With respect to these potential benefits, we can summarize the self-alignment process as follows: The self-alignment process extends our multi-modal alignment approach to scenarios involving a single modality. We select two pre-trained models for the same modality, typically with one serving as a fixed reference (e.g., CLIP-ViP) and the other as the model to be aligned (e.g., OMNIVORE). We then apply the contrastive learning framework outlined in Section 4.1, using the InfoNCE loss to align the representations of the two models. By exploring self-alignment, we aim to uncover the potential benefits of leveraging the diversity and complementary nature of representations within a modality and to investigate whether this approach leads to superior text alignment in terms of differentiating between classes.

4.2 DATASET CONSTRUCTION

4.2.1 MULTI-MODAL DATASET

To manage and preprocess our dataset efficiently, we designed a custom dataset class capable of handling multiple data modalities. Each dataset sample is a dictionary containing data tensors for all initialized modalities along with the corresponding label. For each modality, we sample twelve frames at uniform intervals. Differing frame rates across modalities have been unified during clip extraction for the DAA dataset. Depth and IR single-channel inputs are repeated across three channels. Skeleton 3D joint data is normalized in 3D and treated as image data, where each 3D joint represents a "patch" of an image. For the DAA dataset, we extract all clips from the source videos for RGB, IR, and depth modalities. The frame rate for the newly created clips is fixed at 15 frames per second, while the resolution of the source videos is maintained. The clips are saved with unique identifiers that include participant ID, file ID, frame start, frame end, annotation ID, and chunk ID. In the case of skeleton data, we handle it differently for the DAA and NTU datasets. For the DAA dataset, we reorder the joints to match the order of the NTU dataset. Additionally, we artificially compute several joint positions to ensure compatibility with the NTU dataset format. The middle of the spine is calculated as the midpoint between the 'midHip' and 'neck' joints. The head position is approximated using the 'nose' joint. The left and right foot positions are computed as the average of the respective 'BigToe' and 'Heel' joints. For the left and right hands, we use the 'lWrist' and 'rWrist' joints as placeholders. These artificially computed joints help maintain consistency in the joint representation across both datasets. The skeleton data is then loaded, reordered, and saved as NumPy arrays.

For the NTU dataset, we decided to use masked depth maps from the NTU dataset, primarily for practical reasons related to data management. The use of full-depth maps would require about 886GB of storage, while masked depth maps reduce this requirement to roughly 83GB. This significant reduction in data size is crucial for more efficient processing and management, especially given computational constraints. However, it is important to acknowledge that this choice might introduce some differences in the results when compared to using full depth maps. Masked depth maps focus on specific regions of interest—primarily human figures—and ignore the rest, which could potentially skew the outcomes by omitting contextual depth information that might also be relevant for classification performance.

During the data loading process, we apply various transformations to the skeleton data, such as random rotation, scaling, and Gaussian noise, to augment the data and improve the model's robustness. We also handle cases where the number of frames in the skeleton data is less than the required frame count by repeating the last frame. The base experiments do not

use augmentations. For the final best combinations, augmentations are applied: for DAA, we use random center cropping in a range of 1 to 0.8, and for NTU, we use random center cropping in a range of 0.8 to 0.4. For both datasets, random horizontal flipping with a probability of 0.5 is deployed. During evaluation (validation and test), we use the average range for cropping. All frames are resized to 224x224.

4.2.2 CLIP-ViP RGB-TEXT DATASET

To leverage the CLIP-ViP pipeline for fine-tuning, we decided to use the available CLIP-ViP framework instead of recreating the procedure from scratch. This allows us to efficiently pair videos with text descriptions, which is essential for the CLIP-ViP approach. The NTU and DAA datasets do not come with individual video descriptions; they only provide class names that are consistent across all samples of a given class. While using these class names alone can serve as a baseline for extracting representations, we enhanced this setup by creating more descriptive scene-based text variants.

For the baseline dataset, we used the provided class names directly. Additionally, we created an enhanced version by extending these class names into detailed scene descriptions. To generate these descriptions, we used gpt-4-1106-preview [Ope24] via the ChatGPT interface. For the NTU dataset, we used the following prompt:

“Extend each description to be more descriptive of an actual scene that is unfolding. These are classes from the NTU dataset, where the videos showcase the featured actions in an indoor industrial or office setting. Please conceive realistic scene descriptions for each class. [List of NTU class names]”

For the DAA dataset, we used a similar prompt tailored to its context:

“Extend each description to be more descriptive of an actual scene that is unfolding. These are classes from the DAA dataset, where the videos showcase everyday actions within a vehicle. Please conceive realistic scene descriptions for each class. [List of DAA class names]”

In addition to using a single description, we also created variants of the dataset with six different variants of the scene descriptions and six different synonyms for the class descriptions. For each set of texts, we generated one additional variant at a time and reviewed the results manually to ensure quality and contextual accuracy. This process was repeated until we had six different descriptive variants and six sets of synonyms for each class description. The following prompts were used iteratively to create these variations:

Prompt for Class Names with Synonyms

”Generate a list of synonyms for the following class names. Ensure each synonym is unique yet contextually accurate. [List of class names]”

Prompt for Sentence Variants

”Generate a different variant of the following class names’ descriptions. Ensure each variant is unique yet contextually accurate. [List of class name sentences]”

By creating these extended and varied descriptions, we aim to provide richer textual context for the videos, enhancing the CLIP-ViP model’s ability to learn comprehensive and robust representations. Overall, we end up with four different variants:

- **Words:** The original class names.
- **Sentences:** The extended class descriptions.
- **Variants:** The varied sentences.
- **Synonyms:** The six synonyms of the original class names.

4.2.3 GENERATING A RGB-TEXT DATASET FROM RGB VIDEOS

The recent advancements in vision-language models, such as LLaVA [LLLL23, LLWL24], have presented new opportunities to create text descriptions for RGB videos, specifically targeting the individual frames within these videos. These generated descriptions can then serve as a valuable dataset for contrastive learning objectives.

In this section, we introduce our method for utilizing the LLaVA model to generate captions for RGB video frames to enhance the development of multi-modal detectors across different modalities. We employ the LLaVA-v1.6-34b model [LLWL23].

To create the dataset, we begin with RGB video clips and extract frames from three key points within each clip: one-third, the midpoint, and two-thirds of the way through the clip. This ensures a representation of different stages of the video’s action. Each of these frames is then independently processed using the LLaVA model with the following prompt 3 different times with a temperature setting of 0.5:

”[Frame] Describe the human action in this frame in just one sentence.”

This procedure generates nine captions for each clip, which, when aggregated across the dataset, could provide a robust textual depiction of the visual content. The resulting dataset, includes these multiple descriptions per clip, forming a rich foundation for the RGB-text fine-tuning phase in our multi-modal learning framework. While this dataset generation

technique allows for scalability and reduces the need for manual annotation, it is worth noting that the primary objective of this approach is to establish a proof of concept. The focus at this stage is not necessarily on perfecting the method but on validating the feasibility of using automated caption generation for enhancing our CMRVA framework. Future efforts will likely explore refining these techniques and assessing their effectiveness in more controlled studies to better understand and enhance the accuracy and utility of the generated datasets for multi-modal applications.

5 EXPERIMENTAL SETUP

5.1 OVERALL SETUP DESCRIPTIONS

This section details the common phases shared across all experiments, serving as a foundational reference to streamline subsequent descriptions. The phases include the fine-tuning stage for CLIP-ViP (FT), the alignment phase (AP), and the evaluation phase (EP). The evaluation phase further divides into two distinct main procedures: linear evaluation protocol (LEP) and retrieval (RT). We also briefly introduce k-nearest neighbor (kNN) as an additional evaluation procedure.

5.1.1 FINE-TUNING CLIP-ViP

To fine-tune CLIP-ViP on our datasets, we adopt the fine-tuning infrastructure provided by the authors of CLIP-ViP [XSL⁺22], with slight modifications to work without Apex and Horovod¹ due to compatibility issues with our current environment. The fine-tuning process uses a set of hyperparameters originally employed by the authors for fine-tuning on MSRVT^{TT} [XMYR16] as inferred from their published configuration files on GitHub. These include using 12 frames for both training and testing, as well as input resolutions of 224×224 . The maximum text length is set to 77 tokens, and the CLIP ViT-B/16 model is used as the backbone. The CLIP vision encoder is augmented with 3 ViPs (Video Proxy Tokens), which have a temporal size of 12. For training, we use a batch size of 32. The AdamW optimizer is employed with betas of (0.9, 0.98), a learning rate of $1e-6$, and a weight decay of 0.1. The learning rate is warmed up for 1% of the total training steps and then decayed using a cosine schedule. The model is trained for 10 epochs, except specified differently in a given experiment. The fine-tuning process employs a modified learning objective called NCELearnableTempDSLloss, which differs from the InfoNCE loss used in our alignment phase. This loss function learns a temperature parameter to scale the logits, with the logit scale initialized to 4.60. It applies softmax operations to the rows and columns of the similarity matrix separately, and then multiplies the original similarity matrix element-wise with these softmax outputs. The cross-entropy loss is then computed on this modified similarity matrix. In contrast,

¹Apex and Horovod are optimization frameworks

the InfoNCE loss as introduced in section 4.1 uses a fixed temperature hyperparameter and directly computes the cross-entropy on the similarity matrix without these additional softmax and multiplication steps. Data augmentations, including resizing frames to 224×224 , random center cropping, and horizontal flipping, are applied to the RGB videos in the CLIP-ViP dataset only for the final best model combinations. The random center cropping range is 1 to 0.8 for the DAA dataset and 0.8 to 0.4 for the NTU dataset. Both datasets use random horizontal flipping with a probability of 0.5. Temporal jittering is applied by default during the frame sampling process to introduce randomness in the selection of video frames from various points in time. This is achieved through interval-based sampling, where the video is segmented into intervals, and frames are randomly selected within these intervals. This method helps capture a broader range of temporal variations, enhancing the model’s robustness and reducing its tendency to overfit specific frame sequences.

5.1.2 ALIGNMENT PHASE (AP)

In our multi-modal action recognition pipeline, the alignment phase plays a crucial role in learning a unified embedding space that aligns the representations from different modalities. By bringing the embeddings from various modalities closer together, our model can effectively leverage the complementary information provided by each modality, leading to improved performance and generalization. Our alignment phase requires at least two encoders, each processing its data from the same or a different modality. In this study, we consider four modalities for the AP: RGB, infrared (IR), depth, and skeleton. All experiments include the RGB modality, with its encoder weights either frozen or trainable depending on the specific setup. This design choice allows us to control the number of experiments while still investigating the impact of different alignment strategies. Unless specified otherwise, the encoders for each modality are initialized with their respective pre-trained weights, as detailed in the background Sections for each model (Section 3.3). In cases where an experiment initializes the encoders from scratch, it is explicitly mentioned in the context of that specific experiment. During the alignment phase, we optimize the InfoNCE loss as introduced in section 4.1 to encourage the embeddings of corresponding samples from different modalities to be close to each other while pushing apart the embeddings of non-corresponding samples. We make use of our proposed extension of the InfoNCE loss to handle all pairwise combinations of modalities in an AP with more than two encoders, as described in Equation (4.3), to simultaneously align the embeddings from all modalities. The temperature parameter in the InfoNCE loss is set to 0.1. For the alignment phase, we use a batch size of 32 and sample 12 frames from each video. The Adam optimizer is employed with default beta values of (0.9, 0.999) and a learning rate of 0.0001. Gradients are accumulated over 2 steps. The learning

rate scheduler is configured as a step scheduler with a step size of 4 and a gamma value of 0.1. The model is trained for 10 epochs. The base experiments do not use augmentations. For the final best combinations, augmentations are applied: for DAA, we use random center cropping in a range of 1 to 0.8, and for NTU, we use random center cropping in a range of 0.8 to 0.4. For both datasets, random horizontal flipping with a probability of 0.5 is deployed. During evaluation (validation and test), we use the average range for cropping. All frames are resized to 224x224.

5.1.3 EVALUATION PHASE

LINEAR EVALUATION PROTOCOL (LEP)

The Linear Evaluation Protocol (LEP) serves as a proxy evaluation method to assess the quality of the learned representations. In the context of multi-modal action recognition, LEP assesses how well the aligned embeddings from different modalities can be used to discriminate between different action classes. During LEP, a linear classifier is trained on top of the frozen embeddings extracted from the encoders aligned in the alignment phase. The linear classifier consists of a single linear layer that maps from the embedding dimension to the number of classes. The training and validation processes use the extracted embeddings and corresponding labels for each modality. For the DAA dataset, balanced cross-entropy loss is employed to account for the significant class imbalance present in the data. For the NTU dataset, standard cross-entropy loss is used. By extracting the embeddings beforehand and only training the linear classifier during LEP, the computationally expensive forward pass through the encoders is performed only once during the embedding extraction phase. The LEP evaluation provides insights into how informative the learned representations are for the downstream task of action classification. High performance in this evaluation suggests that the aligned embeddings capture meaningful and discriminative information relevant to distinguishing between different action classes.

RETRIEVAL (RT)

The retrieval evaluation phase assesses the performance of the trained model in retrieving relevant text descriptions that do represent the actual classes for our datasets given visual data. This evaluation is based on the zero-shot retrieval capabilities of the CLIP model [RKH⁺21]. The following is a detailed description of the retrieval process. Let $\mathcal{D} = (\mathbf{x}_i, \mathbf{y}_i)_{i=1}^N$ be a dataset consisting of N samples, where $\mathbf{x}_i$ represents the visual data and $\mathbf{y}_i$ denotes the corresponding text description. First, the text descriptions are tokenized using the CLIP tokenizer, which converts the textual data into a suitable format for the text encoder. The tokenized text

5 Experimental Setup

descriptions are denoted as $\mathbf{t}_i$. The text encoder f_{text} is then applied to the tokenized text descriptions to obtain the text embeddings:

$$\mathbf{e}_i^{\text{text}} = f_{\text{text}}(\mathbf{t}_i) \quad (5.1)$$

where $\mathbf{e}_i^{\text{text}}$ represents the text embedding for the i -th sample. The visual encoder f_{vis} is applied to the visual data to obtain the visual embeddings:

$$\mathbf{e}_i^{\text{vis}} = f_{\text{vis}}(\mathbf{x}_i) \quad (5.2)$$

where $\mathbf{e}_i^{\text{vis}}$ represents the visual embedding for the i -th sample. To measure the similarity between the visual embeddings and the text embeddings, the cosine similarity is computed using the following formula:

$$s_{ij} = \frac{\mathbf{e}_i^{\text{vis}} \cdot \mathbf{e}_j^{\text{text}}}{\|\mathbf{e}_i^{\text{vis}}\| \|\mathbf{e}_j^{\text{text}}\|} \quad (5.3)$$

where s_{ij} represents the cosine similarity between the visual embedding of the i -th sample and the text embedding of the j -th sample, $\cdot$ denotes the dot product, and $\|\cdot\|$ represents the Euclidean norm. The predicted text description for each visual sample is obtained by selecting the text description with the highest cosine similarity. Finally, the retrieval accuracy is computed by comparing the predicted text descriptions with the ground truth labels:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}(\hat{y}_i = y_i) \quad (5.4)$$

where $\mathbb{1}(\cdot)$ is the indicator function that equals 1 if the predicted text description matches the ground truth label, and 0 otherwise. The retrieval evaluation phase provides insights into the model's ability to associate visual data with relevant text descriptions, showcasing its performance in cross-modal retrieval tasks.

K-NEAREST NEIGHBORS (KNN)

In addition to LEP and RT, we employ kNN analysis to assess the quality of our learned representations further. kNN is a non-parametric method used for classification and regression, first introduced by Fix and Hodges [FHJ89]. The kNN algorithm operates on a simple principle: It assigns a data point based on the dominating class of its k-nearest neighbors [SAR23]:

1. Choose a value for k (the number of neighbors to consider). In our evaluations, we use $k=5$.
2. For a given test point:
 - a) Calculate the distance (Cosine Distance for our use case) between the test point and all training points.
 - b) Select the k training points with the smallest distances to the test point.
 - c) Assign the class label based on the majority class among these k neighbors.

The kNN algorithm is particularly valuable for analyzing complex datasets with irregular decision boundaries due to its sensitivity to local data structures [PET09]. Consequently, kNN helps us analyze the structure and separability of the embedding space created by our self-alignment process (see sections 4.1 and 5.2.4). By examining how well kNN can classify points in this space, we gain insights into how effectively our alignment has created distinct, separable clusters for different classes. This complements our LEP and RT evaluations by providing a different perspective on the discriminative power of our learned representations.

5.2 IMPROVING ALIGNMENT

5.2.1 ONE-WAY VS. MUTUAL ADAPTABILITY

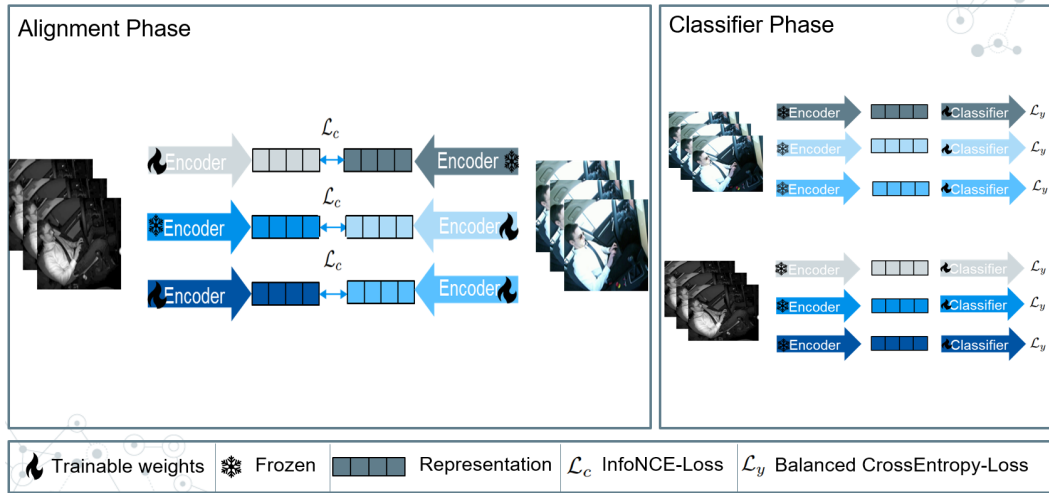


Figure 5.1: One-Way vs Mutual Adaptability: Illustration of comparative Framework for Encoder Adaptability

In the context of multi-modal learning for action recognition, the choice between "binding" the visual encoder to a frozen RGB model and allowing all weights to update during

cross-modal alignment is a critical consideration. The authors of Imagebind [GENL+23] argue that using a strong RGB encoder as a base to "bind" to offers several advantages. Firstly, keeping the weights frozen during the alignment process roughly halves the GPU memory requirements for pairwise alignments. Secondly, it guides the learned embeddings in a direction that has been proven effective for differentiating between different instances. While these arguments are compelling, it is essential to validate whether this approach holds true in action recognition and whether "binding to RGB" outperforms mutual adaptability. This experiment compares one-way and mutual adaptability in our alignment setup. One-way adaptability freezes RGB encoder weights from CLIP-ViP, updating only the other modality's encoder. This reduces memory usage but may limit adaptation to modality-specific features. Mutual adaptability allows both encoders to update, providing more flexibility but increasing computational requirements. We expect faster loss reduction in mutual adaptability but critically analyze whether this correlates with improved downstream performance via the Linear Evaluation Protocol (LEP). The experiment focuses on pairwise alignments between RGB and IR, depth, and skeleton modalities using CLIP-ViP. Two settings are compared for each alignment: one-way (F / T) and mutual (T / T) adaptability. The RGB encoder is fine-tuned on the specific dataset split. Target modality encoders are OMNIVORE for IR and depth, and CLIP-ViP for skeleton. Performance is evaluated using LEP, measuring the accuracy of a linear classifier trained on frozen embeddings of aligned encoders. The experiment also analyzes alignment loss curves to understand learning dynamics and implications for downstream tasks. We hypothesize that binding to RGB will match or outperform mutual adaptation for most modalities and tasks while being more computationally efficient. This assumes the pre-trained RGB encoder provides a strong foundation and stabilizes learning. However, mutual adaptation may excel when the other modality significantly differs from RGB, allowing more flexibility in capturing modality-specific features.

5.2.2 PAIRWISE VS. FULLY-INTEGRATED ALIGNMENT

In our study's contrastive learning approach, we align additional modalities to a frozen RGB encoder, which serves as a strong anchor. This method, however, raises an important question: Can the learning process effectively exploit complementary information from other modalities when the RGB encoder remains fixed? To address this challenge, we propose a fully integrated alignment approach. In this method, the encoders for three additional modalities – depth, IR, and skeleton – are made trainable, while the RGB encoder remains frozen. This setup allows the model to potentially learn and leverage complementary information from these modalities during the alignment process. Our investigation compares two distinct alignment strategies:

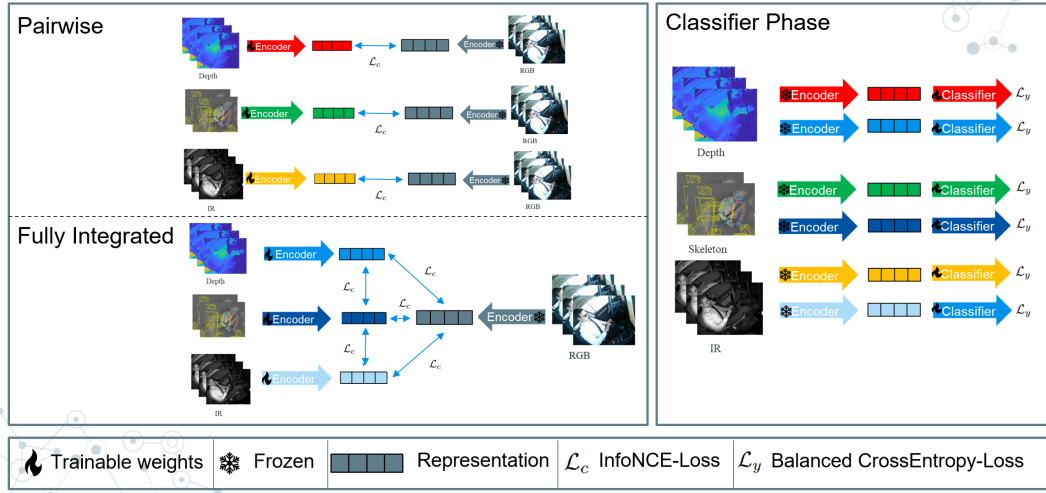


Figure 5.2: Comparison of the alignment and classifier phases in pairwise and fully integrated alignment approaches. In pairwise alignment, each modality is aligned individually with the frozen RGB encoder using the InfoNCE loss. In fully integrated alignment, the InfoNCE loss is applied to all pairs of modalities, encouraging the embeddings from depth, IR, skeleton, and RGB to be aligned simultaneously. The encoders for depth, IR, and skeleton are trainable, while the RGB encoder remains frozen.

- **Pairwise Alignment:** In this approach, each modality (IR, depth, skeleton) is aligned individually with the frozen RGB encoder using its own separate encoder. Contrastive learning is applied to each modality-RGB pair. This setting aims to establish a baseline.
- **Fully Integrated Alignment:** This approach utilizes our InfoNCE extension introduced in section 4.1. Here, the RGB encoder remains frozen while the encoders for depth, IR, and skeleton are trainable. By simultaneously aligning all modalities, this strategy seeks to leverage their combined information, potentially yielding more comprehensive and informative representations.

To gain a nuanced understanding of these approaches, we conduct our experiments using both the base CLIP-ViP version and a version fine-tuned on the specific dataset. This dual approach allows us to observe any differences in behavior between the two versions and assess the impact of dataset-specific fine-tuning on the alignment process. The primary objective of this experiment is to determine whether the fully integrated alignment approach outperforms pairwise alignment in terms of capturing complementary information from different modalities. To evaluate the performance of these approaches, we employ a Linear Evaluation Protocol (LEP) on downstream tasks, which will provide quantitative insights into their effectiveness. Based on the potential for simultaneous learning from multiple modalities, we hypothesize that the fully integrated alignment will outperform pairwise alignment in the

LEP. This outcome would indicate its superior ability to capture and utilize complementary information from different modalities more effectively, potentially leading to more robust and informative multi-modal representations.

5.2.3 LEVERAGING MODALITY SPECIFIC PRETRAINED ENCODERS

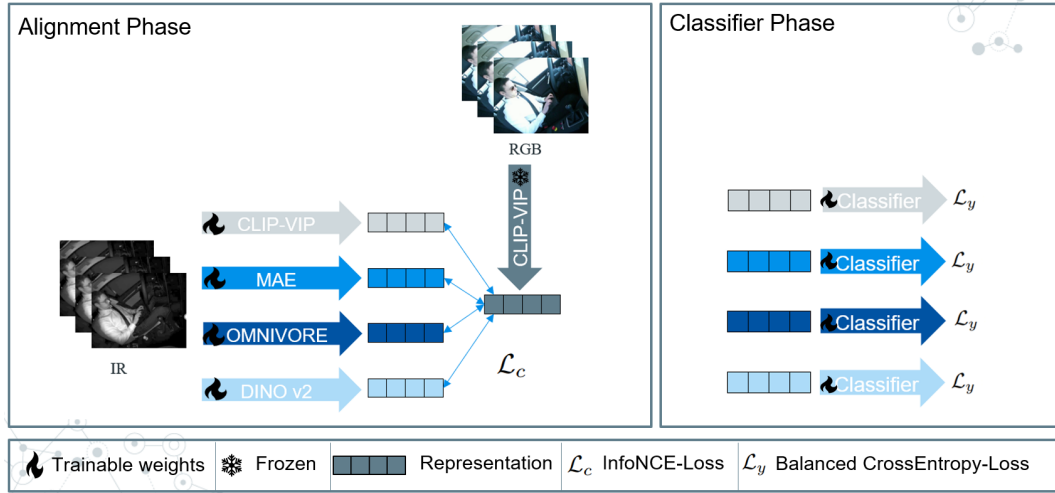


Figure 5.3: Illustration of the analysis for different pre-trained models and their influence on improving the alignment process

Contrastive learning has demonstrated improved performance when trained with large batch sizes and extensive datasets. However, this approach often requires significant computational resources and prolonged training times, especially when the models are trained from scratch. While using pre-trained weights, such as those from CLIP-ViP, can mitigate this issue to some extent, these weights are primarily optimized for the RGB modality and may not be well-suited for other modalities like IR, depth and skeleton data. To address this challenge and improve the efficiency of our multi-modal action recognition framework, we investigate the potential benefits of initializing modality-specific encoders with pre-trained weights that are specifically tailored to their respective modalities. By leveraging modality-specific pre-trained encoders for the IR, depth and skeleton modalities, we aim to accelerate convergence and reduce training time compared to using weights that were initially trained on RGB data. However, it is important to note that this approach introduces additional complexity to the model architecture and relies on the availability of suitable pre-trained models and their publicly accessible weights. In this experiment, we compare two distinct setups:

- **Baseline:** All encoders, including those for depth, IR, and skeleton modalities, are initialized with CLIP-ViP weights, which were originally trained on RGB data.

- Pre-trained: The RGB encoder is initialized with CLIP-ViP weights, while the encoders for the target modalities (depth, IR, and skeleton) are initialized with modality-specific pre-trained models, such as Omnivore, DINO v2, TAHAR, and MAE.

The pre-trained models are carefully selected based on their performance and the availability of their weights. A detailed description of the pre-trained models and their respective strengths is provided in 3.3. It is worth noting that the MAE encoders are self-trained for each specific dataset and split, with details on their training process and hyperparameters provided in Appendix 1. We hypothesize that using modality-specific pre-trained encoders will lead to faster convergence and higher performance compared to using weights initially trained on RGB data for all encoders. However, the magnitude of these gains may vary depending on the specific pre-trained model used for each modality. The primary objective of this experiment is to quantify the efficiency gains achieved by leveraging modality-specific pre-trained encoders in the target modalities while using CLIP-ViP for the RGB modality. By evaluating the impact of modality-specific pre-trained encoders on convergence speed and performance, we aim to determine the effectiveness of this approach in the context of contrastive learning for action recognition, compared to using weights initially trained on RGB data for all modalities. The results will provide valuable insights into the trade-off between the added complexity of using modality-specific pre-trained models and the potential savings in training time and computational resources.

5.2.4 SELF-ALIGNMENT

Building upon our previously introduced concept of self-alignment (see section 4.1), this section details the experimental setup and evaluation methods used to assess its effectiveness in the context of multi-modal action recognition. As established earlier, self-alignment aims to leverage the complementary strengths of different pre-trained models within a single modality, potentially reducing our dependence on annotated data. Our experiments focus specifically on aligning RGB representations using the CLIP-ViP and OMNIVORE models. To perform self-alignment, we freeze the CLIP-ViP model and align the OMNIVORE model to it. We conduct the alignment using both the base version and the fine-tuned version of CLIP-ViP. By keeping CLIP-ViP fixed, we can leverage its text encoder for retrieval tasks. This alignment setup allows us to investigate the impact of self-alignment on the quality and effectiveness of the RGB representations learned by the newly aligned OMNIVORE encoder. The main experiments focus on evaluating the self-aligned representations using our main evaluation procedures LEP and RT. To further understand the factors influencing the quality of the self-aligned representations, we conduct several ablation experiments. We perform

k-nearest neighbors (kNN) analysis on the DAA dataset to gain additional insights into the embedding space of the self-aligned models. This analysis helps us investigate the structure and separability of the embedding space, providing a more comprehensive understanding of the self-aligned representations. In addition to the kNN analysis, we explore the impact of the quality of the OMNIVORE weights on the self-alignment process. We initialize the OMNIVORE model from scratch and align it with the base CLIP-ViP model. By comparing the results with the main experiments, we can assess the role of the OMNIVORE weights in the quality of the self-aligned representations. To investigate the impact of the loss function, we experiment with an alternative loss function, Mean Squared Error (MSE), which measures the average squared difference between the estimated values and the actual value. By using MSE, we encourage the embeddings to strive towards the same numerical values, essentially assessing the role of the off-diagonal value minimization in the cosine similarity matrix of the InfoNCE loss. This experiment aims to provide insights into the importance of the contrastive loss in the quality of the self-aligned representations. Lastly, we visualize and analyze the embedding space of the self-aligned models. We plot the cosine distances between the self-aligned RGB embeddings and the corresponding text embeddings. Additionally, we compute the ratio between the cosine distance to the correct text embedding and the average cosine distance to the remaining incorrect text embeddings. By examining the distribution and separability of the embedding space based on these metrics, we can assess the discriminative power of the self-aligned representations and gain insights into their effectiveness. Through these main experiments and ablation experiments, we aim to validate our hypothesis that self-alignment of RGB representations will lead to improved performance of the target encoder (OMNIVORE) compared to the anchor encoder (CLIP-ViP) in both retrieval tasks and linear evaluation protocol. This hypothesis suggests that the self-alignment process allows the OMNIVORE model to effectively learn from and potentially surpass the performance of the CLIP-ViP model, even though CLIP-ViP serves as the fixed reference during alignment. If confirmed, this would demonstrate the power of self-alignment in transferring knowledge between models and potentially improving upon strong pre-trained representations without the need for additional labeled data.

5.3 APPLYING ALIGNMENT: PRACTICAL BENEFITS AND USES

5.3.1 CROSS DATASET AND CROSS VIEW GENERALIZABILITY

The cross-dataset and cross-view evaluations in this section aim to investigate the generalizability and transferability of the aligned multi-modal embeddings obtained using our proposed methods. While the transferability of CLIP weights to video has been demonstrated by

CLIP-ViP and other previous works, our focus here is on assessing how well our alignment techniques enable the learning of robust and generalizable representations across different datasets and viewpoints, especially for the aligned target modalities.

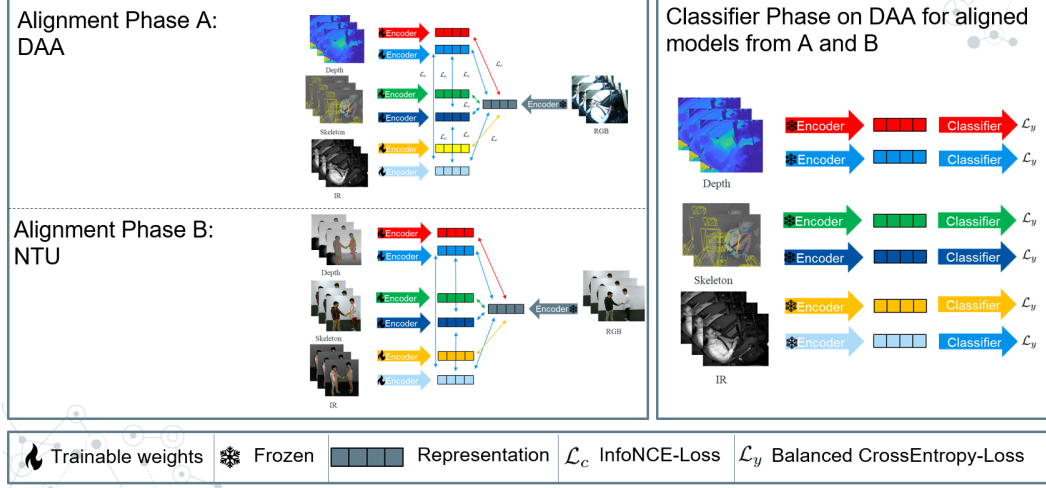


Figure 5.4: Overview of the cross-dataset and cross-view evaluation setup. In Alignment Phase A, pairwise and fully integrated alignments are performed on the DAA dataset. In Alignment Phase B, the alignments are conducted on the NTU dataset. The classifier phase evaluates the generalizability of the aligned models by training classifiers on the DAA dataset for models aligned in both Phase A and Phase B.

The NTU and DAA datasets provide a unique opportunity for this evaluation, as they differ in size, balance, and domain-specific characteristics. By aligning the additional visual modalities (depth, infrared, and skeleton) to the fine-tuned CLIP-ViP representations on the NTU dataset and evaluating their transferability to the DAA dataset, we can gain insights into the effectiveness of our alignment methods in capturing meaningful action recognition features that generalize across datasets. Furthermore, the multiple infrared views available in the DAA dataset allow us to assess the robustness of the aligned embeddings to viewpoint changes. By training on one view and testing on others, we can evaluate the view-invariance of the learned representations, which is a crucial aspect of generalizability in real-world applications. If our alignment methods successfully capture the essential features of human actions, we would expect the learned representations to be transferable across different viewpoints, enabling the recognition of actions regardless of the camera angle or perspective.

The differences in size, balance, and domain-specific characteristics between the NTU and DAA datasets raise an important question: How do these factors affect the generalizability and transferability of our aligned multi-modal embeddings? Intuitively, we might expect that alignments learned on a larger and more balanced dataset, such as NTU, would yield more ro-

bust and transferable representations compared to those learned on a smaller and imbalanced dataset like DAA. This leads us to our hypothesis: the encoders aligned on the larger, balanced NTU dataset will outperform those aligned on the smaller, imbalanced DAA dataset in the Linear Evaluation Protocol (LEP). Our experiments for this section can be summarized:

1. **Cross-Dataset:** We perform pairwise and fully integrated alignment on both NTU and DAA datasets using the base version and the fine-tuned version of CLIP-ViP. For all aligned models, we conduct the Linear Evaluation Protocol (LEP) on the DAA dataset and compare the results. We use only CLIP-ViP for target modalities in this evaluation.
2. **Cross-View:** We align RGB (using fine-tuned CLIP-ViP) and the Kinect IR modality (using OMNIVORE) on the DAA dataset. We then train a classifier on the embeddings of the Kinect IR modality and evaluate the quality of the OMNIVORE checkpoint with the trained classifier on the five additional IR views available in the DAA dataset (see section 2.6).

By conducting these experiments, we aim to gain a deeper understanding of how our alignment methods perform in terms of generalizability and transferability across datasets and viewpoints. The findings from these evaluations will provide valuable insights into the strengths and limitations of our proposed techniques in learning robust multi-modal representations for action recognition in different contexts. Successful cross-view transferability would support the broader goal of our thesis, demonstrating the effectiveness of our approach in learning view-invariant representations.

5.3.2 RETRIEVAL AND LEP

The CLIP-ViP model’s fine-tuning pipeline offers a practical advantage by enabling the use of the language encoder directly for retrieval tasks, without the need for additional supervised training. This is made possible by aligning the visual and language encoders in a shared embedding space, where the visual encoder is frozen (“bound”) to the RGB modality during the alignment process. The resulting alignment allows for efficient and flexible retrieval, as the language encoder can be used to compare text and video embeddings directly. However, it is crucial to investigate the performance gap between using the language encoder for retrieval and employing a trained linear classifier. While the direct use of the language encoder offers practical benefits, it may not achieve the same level of performance as a dedicated classifier trained with labeled data. This raises the question of whether the efficiency and flexibility offered by the language encoder-based retrieval come at the cost of reduced performance compared to a supervised approach. To address this question, we set the objective of comparing

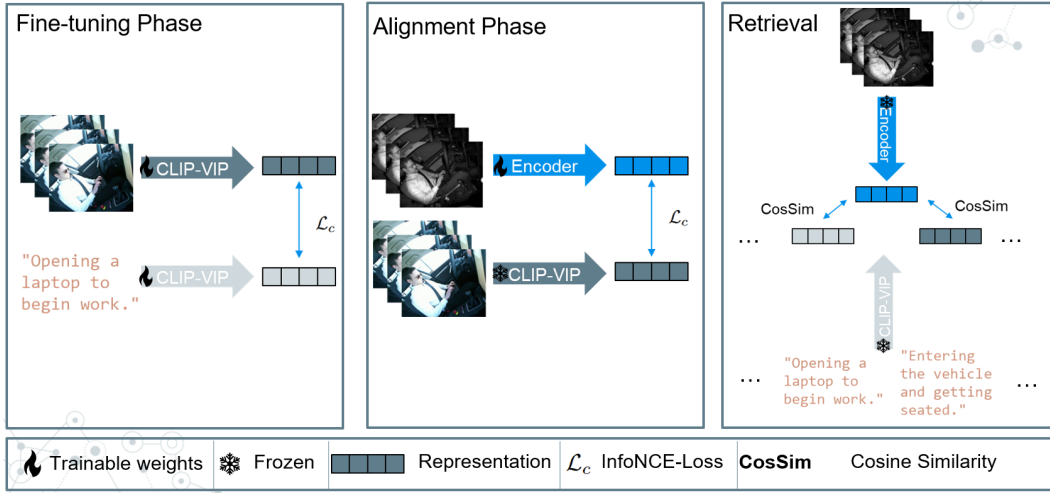


Figure 5.5: Overview of the retrieval evaluation process. After aligning the target modality encoders (IR, depth, skeleton) to the fine-tuned CLIP-ViP RGB encoder, retrieval performance using the CLIP-ViP text encoder is compared against a trained linear evaluation protocol (LEP) classifier.

the performance of the CLIP-ViP language encoder for retrieval tasks versus LEP. By conducting this comparative analysis, we aim to assess the trade-off between the practical benefits of using the language encoder directly and the potential performance gains obtained by employing a supervised LEP. The procedure for this comparison involves performing pairwise alignments between all possible target modalities using the best pre-trained model for each modality. Specifically, we align the IR and depth modalities using the OMNIVORE model, while the skeleton modality is aligned using a modified CLIP-ViP model. These alignments are performed against a fine-tuned CLIP-ViP version serving as the RGB reference. Once the alignments are complete, we utilize the text encoder of the CLIP-ViP model for retrieval tasks. The retrieval performance is then compared to the results obtained using an LEP trained on the frozen visual embeddings. By evaluating both the retrieval and LEP performance across different modalities and datasets, we can gain insights into the effectiveness of the language encoder-based retrieval approach. We hypothesize that retrieval using the language encoder will be competitive with the LEP for many tasks while being more efficient and flexible. The findings of this comparative analysis will provide valuable insights into the trade-offs between the practical advantages of using the language encoder directly for retrieval and the potential performance gains obtained by employing a supervised LEP. This understanding will inform the design choices for downstream tasks in multi-modal action recognition systems, helping to strike a balance between efficiency, flexibility, and performance considerations.

5.3.3 FUSION PERFORMANCE AND MISSING MODALITY SCENARIO

To assess whether our encoders capture unique, complementary features during alignment, we conduct a fully integrated alignment using the best model combinations from previous experiments. Specifically, we employ a fully integrated alignment with a fine-tuned and frozen RGB CLIP-ViP encoder. Concurrently, we align the IR and Depth modalities using an OMNIVORE encoder and the Skeleton modality with a CLIP-ViP encoder. Post alignment, the evaluation phase (LEP) is modified to include training an attention-based fusion head on the combined outputs of all separate encoders. Each encoder is also evaluated independently.

During the fusion head’s training, we employ the same hyperparameters as those used for the linear classifiers. The input to the fusion head consists of concatenated embeddings from each modality. To enhance the model’s robustness and prevent overreliance on any single modality, during training, we intermittently block embeddings from each modality—effectively setting them to zero. Specifically, each modality has a 10% chance of being blocked for each training sample. This blocking occurs independently for each modality, allowing for scenarios where multiple modalities may be blocked simultaneously.

The rationale behind focusing on fusion performance and scenarios with missing modalities is rooted in the realistic conditions under which human action recognition systems operate. In real-world applications, certain modalities may be unavailable or obscured, for example, due to sensor malfunctions or environmental conditions. Therefore, evaluating the system’s performance under these constraints is critical to ensure its reliability and robustness.

For the final robustness evaluation, we conduct multiple runs on the test set:

1. With all modalities present.
2. With one modality blocked.
3. With two modalities blocked.

This step-by-step reduction in modalities helps us understand the contribution of each modality to the overall performance and the fusion head’s ability to compensate for missing data.

We hypothesize that the fusion performance, when evaluated through an attention-based mechanism, should surpass the performance of any individual encoder assessed with LEP. This is because the fusion mechanism leverages combined features that potentially encapsulate a broader and more diverse data spectrum, leading to a more accurate interpretation and classification of human actions. Therefore, the integrated model should not only perform better with all modalities present but also demonstrate greater resilience when modalities are

missing, compared to separate encoders evaluated individually. This hypothesis will be tested by comparing the robustness evaluations and seeing if the fusion model maintains high accuracy across different scenarios of modality availability.

5.3.4 ZERO-SHOT EVALUATION ON DRIVE&ACT

Zero-shot recognition has emerged as a challenging and practically relevant problem in computer vision, requiring models to accurately classify instances from classes that were not encountered during training. This task is particularly important in real-world applications, such as driver behavior recognition, where the ability to generalize to novel activities is crucial for robust performance. Our multi-modal alignment framework, with its emergent alignment to the text modality, provides a natural avenue for exploring zero-shot evaluation, as the language encoder can be leveraged to retrieve the most relevant unseen class for a given input. To assess the generalizability and robustness of our approach in this context, we conduct a zero-shot evaluation on the DAA dataset, following the protocol introduced by Reiß et al. [RRHS20]. It is worth noting that a zero-shot evaluation protocol has also been proposed for the NTU RGB+D dataset [ZKGB24], focusing specifically on the skeleton modality. However, for the scope of this work, we choose to focus on the DAA dataset and its zero-shot protocol, leaving the exploration of the NTU zero-shot evaluation for future research. The DAA dataset encompasses 34 fine-grained driver activity classes, capturing a diverse range of behaviors within the vehicle cabin. For zero-shot evaluation, these classes are divided into seen and unseen classes, adhering to the 14-10-10 split proposed by Reiß et al. [RRHS20]. In this split, 14 classes are designated as seen classes and used for training the models, while the remaining 20 classes are evenly divided into validation and testing sets, representing the unseen classes for which the models will be evaluated. To mitigate the potential impact of specific class splits on the results, the evaluation is repeated over 10 distinct 14-10-10 splits published by Reiß et al. [RRHS20], and the average top-1 accuracy on the test set is reported. Our experimental pipeline for the zero-shot evaluation on DAA begins with fine-tuning the CLIP-ViP RGB encoder on the 14 seen classes for 5 epochs, utilizing the RGB frames and their associated class names. Subsequently, we perform self-alignment between the fine-tuned CLIP-ViP RGB encoder and an OMNIVORE RGB2 encoder for 10 epochs, enabling the OMNIVORE encoder to align its representations with those of the fine-tuned CLIP-ViP encoder. We then align the OMNIVORE IR encoder to the fine-tuned CLIP-ViP RGB encoder for 10 epochs, bridging the gap between the RGB and IR modalities. Finally, for each test instance from the unseen classes, we compute the cosine similarity between its encoded representation and the language embeddings of the unseen classes, predicting the class with the highest similarity as the output of the model. By following this experimental

setup, we aim to evaluate the effectiveness of our multi-modal alignment approach in generalizing to novel driver behaviors and assessing its potential for open-set recognition scenarios.

5.4 BEYOND STRICT LABELS: LEVERAGING LANGUAGE FOR ALIGNMENT

5.4.1 VARIETY IN DESCRIPTIONS

Building upon our previous alignment experiments, we now investigate how varying degrees of language diversity during fine-tuning affect both RGB encoder performance and the subsequent alignment of target modalities. In earlier experiments, we compared base and fine-tuned versions of our models. This section extends that work by exploring different variations of fine-tuned models, each trained on a distinct level of textual diversity. Our primary objective is to determine whether increased language variety during fine-tuning leads to enhanced RGB encoder performance and improved alignability and performance of target modalities, specifically IR in this case. To this end, we utilize four variants of our CLIP-ViP dataset, each offering a different level of textual richness. These variants range from the original class names (Words), which offer the least diversity, to extended class descriptions (Sentences), multiple varied sentences for each class (Variants), and finally to six synonyms for each original class name (Synonyms), providing the most diversity. The detailed process of creating these dataset variants is described in Section 4.2.2. We conduct separate fine-tuning phases for each variant, saving model checkpoints at 5, 10, and 15 epochs. This approach allows us to track performance evolution and assess potential overfitting. Following fine-tuning, we align an IR encoder (using OMNIVORE) to each of these RGB encoder variants. We then evaluate both the RGB and aligned IR encoders using LEP across all variants and epoch checkpoints. This comprehensive evaluation enables us to analyze how different levels of language diversity during fine-tuning impact RGB encoder performance, whether increased textual variety leads to RGB representations that are more conducive to aligning other modalities and the relationship between fine-tuning duration and alignment quality. We hypothesize that greater language variety, as found in the Variants and Synonyms datasets, will not only improve RGB encoder performance but also facilitate better alignment and performance of the IR encoder. This experiment aims to provide insights into optimizing the fine-tuning process for improved multi-modal alignment.

5.4.2 VISION LANGUAGE MODELS FOR LABEL GENERATION

The recent advancements in vision-language models, such as LLaVA [LLLL23, LLWL24], have opened up new possibilities for generating textual descriptions from visual data. In this section, we investigate the potential of using machine-generated captions from the LLaVA model as substitutes for traditional manual annotations in fine-tuning CLIP-ViP. In a second step we analyse the behavior of this fine-tuned version in our multi-modal alignment framework. We focus on two alignment setups: RGB to IR alignment and RGB to RGB self-alignment. For each alignment setup, we evaluate the performance using RT and LEP. The experiments are conducted with three variations of the anchor RGB encoder:

- Base weights (no fine-tuning)
- RGB encoder fine-tuned on generated captions from LLaVA
- RGB encoder fine-tuned on real manual annotations

The base version of CLIP-ViP, which utilizes weights derived from large-scale training, serves as a control to benchmark the performance enhancements offered by LLaVa-generated captions. All in all the core of our experiment involves fine-tuning the RGB encoder with video clips from our dataset, annotated with either captions generated by the LLaVA model or real manual annotations (the actual class names). The process of generating these captions with LLaVa is detailed in section 4.2.3. It is important to note that the generated captions may introduce noise, as they might not accurately reflect the actions taking place in the video. After fine-tuning the RGB encoder, we align the IR modality encoder to the updated RGB encoder in the RGB to IR alignment setup, and in the RGB to RGB self-alignment setup, we align another RGB encoder to the fine-tuned RGB encoder. Both the IR and RGB encoders are based on the OMNIVORE model. To evaluate the effectiveness of this approach, we employ RT and LEP. In the retrieval tasks, we assess the performance of the fine-tuned RGB encoder and the aligned IR encoder (or the self-aligned RGB encoder) using the text encoder corresponding to each fine-tuning phase: For the base setting (no fine-tuning), we use the original class names for retrieval. For the fine-tuned settings, we use the text encoder fine-tuned on generated captions or real labels, respectively. This allows us to determine the effectiveness of synthetic labels in facilitating accurate matches from RGB and IR data (or self-aligned RGB data) compared to using real labels or no fine-tuning at all.

We hypothesize that training and aligning the encoders using captions generated by the LLaVA model will:

5 *Experimental Setup*

- For retrieval tasks: Outperform the base version (no fine-tuning) by leveraging the semantic information captured in the generated captions, potentially leading to more accurate matches between RGB and IR data (or self-aligned RGB data).
- For LEP evaluations: Achieve comparable performance to the encoders fine-tuned on real manual annotations, demonstrating the effectiveness of using synthetic labels in learning meaningful representations for downstream tasks.

The combination of artificial label generation using vision-language models with self-alignment may offer potential benefits for action recognition tasks. While the current label generation approach relies on single frames, which is a simplification, this experiment aims to explore whether such a method could yield promising results. If successful, it could motivate further research into leveraging the semantic knowledge captured by these vision-language models to potentially enhance action recognition systems. Additionally, this approach might help reduce reliance on manual labeling methods within our CMRVA framework.

6 RESULTS

6.1 IMPROVING ALIGNMENT

6.1.1 ONE-WAY vs. MUTUAL ADAPTABILITY

To validate the hypothesis that one-way adaptability outperforms mutual adaptability in multi-modal alignment, we conducted experiments comparing the two approaches across the DAA and NTU datasets for RGB, infrared (IR), depth, and skeleton modalities. The results, summarized in Table 6.1, consistently demonstrate that one-way adaptability, where the RGB encoder is frozen and the target modality encoder is trainable, achieves superior performance on downstream tasks as measured by LEP accuracy. Therefore supporting our hypothesis. Notably, for the DAA dataset, one-way adaptability surpassed mutual adaptability by 7-10 percentage points across all modalities. Similar trends emerged in the NTU cross-view and cross-subject splits, with one-way adaptability maintaining a clear advantage.

Table 6.1: One-Way vs. Mutual Adaptability on Pairwise Alignments. Performance is evaluated for CLIP-VIP Fine-Tuned (RGB) to OMNIVORE (IR), CLIP-VIP Fine-Tuned (RGB) to OMNIVORE (DEPTH), and CLIP-VIP Fine-Tuned (RGB) to CLIP-VIP (Skeleton).

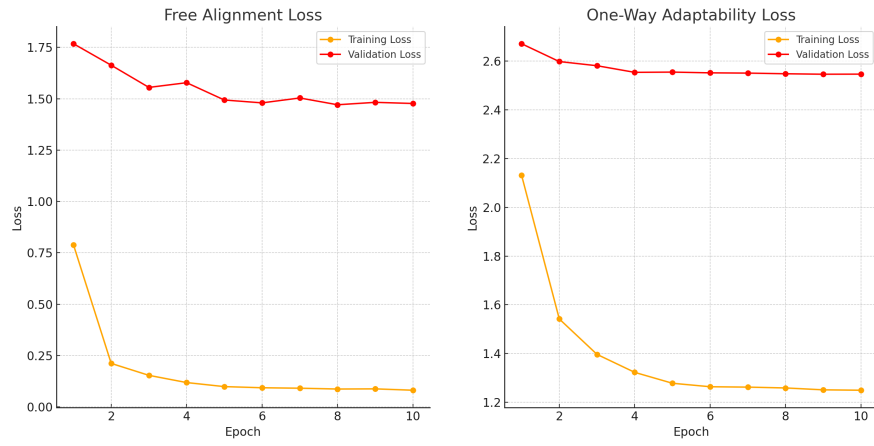
Dataset Split	Freeze Pattern	LEP Accuracy Metrics					
		RGB	IR	RGB	DEPTH	RGB	SKELETON
DAA (avg)	F / T	64.41	57.66	64.40	49.29	63.20	42.03
	T / T	57.44	50.45	46.88	40.07	39.88	32.71
NTU (CV)	F / T	77.48	73.36	77.82	77.67	77.73	71.9
	T / T	59.08	56.29	55.74	57.23	68.69	60.9
NTU (CS)	F / T	76.09	80.94	76.09	80.31	76.06	67.12
	T / T	58.48	59.54	58.07	57.08	65.85	54.67

Interestingly, the validation loss dynamics (Table 6.2 and Figure 6.1) reveal that mutual adaptability achieves lower InfoNCE losses during training, suggesting tighter alignment between modalities. However, this tighter alignment does not translate to improved class separability on downstream tasks. We expect two key factors contributing to one-way adaptability’s superior performance. Firstly, using a strong, fine-tuned RGB encoder as a stable

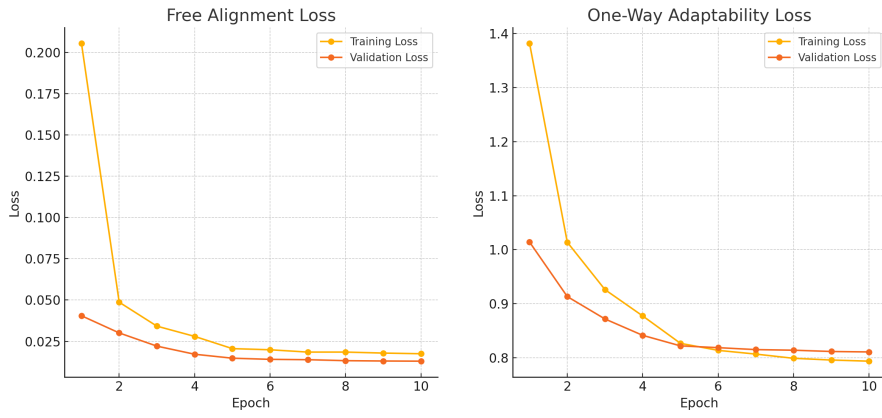
Table 6.2: Validation Loss After Ten Epochs Pairwise Alignment

Freeze Pattern	RGB / IR	RGB / DEPTH	RGB / SKELETON
DAA (avg)			
F / T	2.409	2.235	2.618
T / T	1.466	1.561	2.129
CV			
F / T	0.911	0.985	1.290
T / T	0.018	0.028	0.337
CS			
F / T	0.811	0.881	1.189
T / T	0.029	0.025	0.306

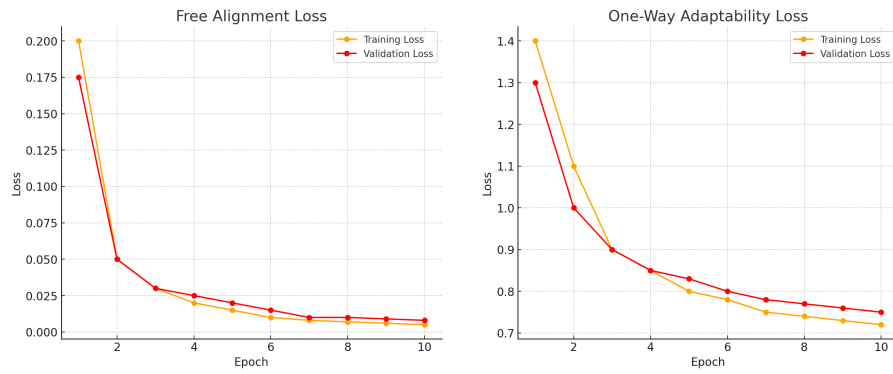
anchor for guiding the alignment process prevents the representations from diverging too far from the well-established and informative features captured by the pre-trained encoder. This stability helps to mitigate representation drift and promotes better class separability, even if the alignment between the modalities during training is not as tight as in the mutual adaptability setting. Secondly, the RGB-text fine-tuning process in CLIP-ViP treats samples as class instances, which could lead to more discriminative representations. In contrast, during the pairwise alignments, each sample is considered its own unique pairing, which may not explicitly encourage the same level of class-based separability. This difference in the training objectives may contribute to the superior downstream performance of the one-way adaptability setting, as the RGB encoder is guided by a more discriminative learning signal. Our findings align with and extend the ImageBind approach [GENL+23] to temporal data from diverse modalities. ImageBind demonstrated the effectiveness of using a frozen CLIP image encoder as an anchor for aligning representations from various image-based modalities. Our experiments show that this one-way adaptability strategy is also effective for temporal data, such as videos and sequences, across RGB, infrared, depth, and skeleton modalities. This suggests that the ImageBind methodology can be generalized to a wider range of data types and modalities, providing a powerful framework for multi-modal representation learning. Moreover, the one-way framework offers additional benefits. By freezing the RGB encoder, we reduce computational costs and memory requirements during training. Furthermore, the frozen RGB encoder maintains its alignment to the text modality, enabling zero-shot classification via language queries without additional fine-tuning. In conclusion, our experiments provide compelling evidence for the superiority of one-way adaptability in our multi-modal alignment framework. By leveraging a fixed RGB encoder as an anchor, we can learn more



(a) DAA (split 0)



(b) NTU (CS)



(c) NTU (CV)

Figure 6.1: Comparison of training and validation loss across different dataset splits. Free alignment denotes mutual adaptability

discriminative and separable representations while also in theory enabling emergent alignment to language.

6.1.2 PAIRWISE VS. FULLY INTEGRATED ALIGNMENT

We introduce a novel fully integrated alignment strategy that simultaneously aligns representations from multiple modalities, by extending the InfoNCE loss as outlined in Section 4.1, aiming to capture complementary information more effectively compared to pairwise alignment. To evaluate the performance of these approaches, we conduct experiments using LEP on downstream tasks across the NTU60 (CS and CV) and DAA datasets. The results for the NTU120 dataset (CS120 and CV120) are reported in Appendix 4, as they exhibit similar trends to the NTU60 dataset, differing only in magnitude. The experimental results, presented in Tables 6.3-6.5, reveal several key findings. Firstly, fully integrated alignment demonstrates superior performance compared to pairwise alignment when using the base CLIP-ViP model. This observation is consistent across all datasets, with the fully integrated approach (R I D S) achieving higher accuracies for the majority of modalities. We hypothesize that the base version of CLIP-ViP groups samples close together in the embedding space, and the fully integrated alignment strategy effectively spreads them out, leading to improved performance. In stark contrast, pairwise alignment significantly outperforms fully integrated alignment when using the fine-tuned CLIP-ViP model. For instance, in the CS dataset (Table 6.3), fine-tuned pairwise alignment (R I) achieves an impressive 54.37% accuracy for, while the fully integrated approach (R I D S) only reaches 35.85% for the IR modality. This trend is consistent across all datasets and splits, with the magnitude of the performance gap varying slightly. We speculate that if the embeddings are already more spread out after fine-tuning, the fully integrated alignment may face difficulties or even break down entirely. The low accuracies observed in the alignment towards a fine-tuned CLIP-ViP, especially for the CV dataset (Table 6.4), are quite drastic in that regard. In this case, the fine-tuning process might have reached an optimum, in which the three trainable encoders might group their embeddings together in a suboptimal configuration without moving closer to the frozen RGB encoder, leading to a significant drop in performance compared to pairwise alignment. Another notable observation is the varying performance of individual modalities across datasets and alignment strategies. The RGB modality generally achieves the highest accuracies, followed by skeleton, IR, and depth. However, the fully integrated approach often narrows the performance gap between modalities when using the base model, highlighting its ability to better leverage complementary information from multiple sources in this setting. In conclusion, our experiments demonstrate that the proposed fully integrated alignment strategy is more effective at capturing complementary information from multiple modalities

compared to pairwise alignment when using a base model with foundational representations. This suggests that fully integrated alignment can help improve representation quality in cases where RGB-text fine-tuning is not possible. However, for fine-tuned models, pairwise alignment consistently outperforms fully integrated alignment, indicating that the latter may not be suitable for models that have undergone extensive dataset-specific refinement. These findings highlight the importance of carefully considering the nature of the base representations and the alignment strategy when designing multi-modal learning systems. Further research is needed to better understand the factors contributing to the divergent behavior of fully integrated alignment in base and fine-tuned settings.

Table 6.3: LEP Accuracies for CS on NTU

Frozen RGB Model	Aligned Modalities	Accuracies			
		R	I	D	S
CLIP-VIP base	R I		55.66	-	-
	R D	53.84	-	31.33	-
	R S		-	-	55.6
	R I D S		60.21	46.62	58.73
CLIP-VIP fine tuned	R I		54.37	-	-
	R D	76.04	-	41.68	-
	R S		-	-	67.12
	R I D S		35.85	48.56	62.37

Table 6.4: LEP Accuracies for CV on NTU

Frozen RGB Model	Aligned Modalities	Accuracies			
		R	I	D	S
CLIP-VIP base	R I		54	-	-
	R D	56.50	-	25.99	-
	R S		-	-	62.56
	R I D S		56.19	49.22	64.22
CLIP-VIP fine tuned	R I		51.7	-	-
	R D	77.63	-	48.75	-
	R S		-	-	71.9
	R I D S		14.9	43.47	66.03

Table 6.5: LEP Accuracies for DAA (avg)

Section	Aligned Modalities	Accuracies (%)			
		R	I	D	S
CLIP-ViP base	R I		19.59	-	-
	R D		-	15.89	-
	R S	47.26	-	-	31.55
	R I D S		23.27	16.44	28.40
CLIP-ViP fine tuned	R I		23.11	-	-
	R D		-	19.41	-
	R S	63.06	-	-	42.03
	R I D S		18.53	10.48	32.85

6.1.3 LEVERAGING MODALITY SPECIFIC PRE-TRAINED ENCODERS

To improve the efficiency and performance of our multi-modal action recognition framework, we investigate the impact of initializing modality encoders with pre-trained weights specifically tailored to their respective modalities. By leveraging modality-specific pre-trained encoders for the depth, infrared (IR), and skeleton modalities, we aim to accelerate convergence and boost downstream performance compared to using weights initially trained on RGB data. Table 6.6 demonstrates that using modality-specific pre-trained encoders generally leads to better downstream performance compared to initializing all encoders with CLIP-ViP weights, which were originally trained on RGB data. We observe the same trends for alignment towards CLIP-ViP base but report the results in Appendix 4 for brevity. Across most datasets, the pre-trained encoders outperform the baseline CLIP-ViP encoders for their respective modalities after alignment.

The OMNIVORE encoder consistently achieves the highest test accuracies for the infrared and depth modalities, ranging from 57.66% on DAA to 80.94% on NTU (CS) for infrared, and 49.29% on DAA to 80.31% on NTU (CS) for depth. These results surpass the performance of the baseline CLIP-ViP encoder and other modality-specific encoders like MAE and DINOv2. While the Masked Autoencoder (MAE) approach, which involves self-supervised pre-training on modality-specific data, demonstrates improved performance compared to the base CLIP-ViP encoder, it does not consistently reach the performance levels of the OMNIVORE encoder. On the DAA dataset, MAE achieves test accuracies of 39.66% and 42.17% for the infrared and depth modalities, respectively, when aligned with a fine-tuned RGB encoder, surpassing the corresponding CLIP-ViP accuracies. The performance gap between MAE and OMNIVORE can be attributed to two factors. Firstly, our MAE

Table 6.6: LEP for different Encoder Models aligned to a frozen fine-tuned CLIP-ViP

Dataset	Modality	Architecture	Test Accuracy	RGB Accuracy
DAA (Average)	Infrared	CLIP-ViP	23.12	63.40
		DINO	10.82	
		DINO (fro)	17.58	
		MAE	39.66	
	Depth	OMNIVORE	57.66	
		CLIP-ViP	19.42	
		MAE	42.17	
	Skeleton	OMNIVORE	49.29	
		CLIP-ViP	42.03	
NTU (CV)	Infrared	TAHAR	39.01	77.82
		CLIP-ViP	51.70	
		DINO (fro)	29.45	
		MAE	61.41	
	Depth	OMNIVORE	73.36	
		CLIP-ViP	48.75	
		MAE	74.91	
	Skeleton	OMNIVORE	77.67	
		CLIP-ViP	71.90	
NTU (CS)	Infrared	TAHAR	68.96	76.08
		CLIP-ViP	54.37	
		DINO (fro)	32.25	
	Depth	OMNIVORE	80.94	
		CLIP-ViP	41.68	
	Skeleton	OMNIVORE	80.31	
NTU 120 (CP)	Infrared	CLIP-ViP	67.12	66.83
		TAHAR	58.57	
	Depth	CLIP-ViP	52.37	
		OMNIVORE	71.70	
	Skeleton	CLIP-ViP	39.61	
		OMNIVORE	67.95	
NTU 120 (CS)	Infrared	CLIP-ViP	59.89	70.87
		TAHAR	53.76	
	Depth	CLIP-ViP	72.55	
		OMNIVORE	76.75	
	Skeleton	CLIP-ViP	54.66	
		OMNIVORE	73.01	
	Skeleton	CLIP-ViP	60.10	
		TAHAR	54.67	

training implementation, as detailed in Appendix 1, may be suboptimal. Secondly, OMNIVORE’s architecture as a universal encoder is specifically designed to capture generalizable features across diverse modalities, as discussed in Section 2.5, potentially giving it an inherent advantage in cross-modal tasks.

Adapting the DINOv2 encoder, pre-trained on image data using self-supervised learning, for the infrared modality shows bad results. In the base configuration, DINOv2 achieves a test accuracy of only 10.82% on the DAA dataset, significantly lower than other architectures. However, when the DINOv2 encoder is frozen during alignment (DINO (fro)), its performance improves to 17.58% on DAA and 29.45% on NTU (CV) with a fine-tuned RGB encoder. Despite these improvements, DINOv2 still lags behind the other modality-specific architectures, possibly due to its primary design for image-level tasks, which may not effectively capture the temporal dynamics in infrared video data. Future work could explore more sophisticated techniques to adapt DINOv2 for video-based tasks, such as incorporating temporal adapters or video proxy tokens, as used in the CLIP-ViP architecture.

An unexpected result is observed for the skeleton modality: the CLIP-ViP encoder, despite not being pre-trained on skeleton data, consistently outperforms the specialized TAHAR encoder across all datasets. This could be attributed to two factors: firstly, TAHAR was trained on significantly more frames on average, while the current pipeline uses only 12 frames; secondly, TAHAR’s significantly smaller model capacity compared to CLIP-ViP may limit its ability to align its skeleton representations with RGB representations. However, it is worth noting that TAHAR’s compact architecture makes it much more efficient in terms of computational resources and memory requirements. In addition to that the performance gap is significantly smaller than the gap observed for different IR and depth models.

All in all the experimental findings largely support the hypothesis that using modality-specific pre-trained encoders leads to faster convergence and higher performance compared to RGB-trained weights. This is evidenced by substantial performance gains in infrared and depth modalities, although results for skeleton data slightly contradict this trend. It’s worth noting that other pre-trained models might yield different outcomes. While integrating pre-trained models into the alignment process introduces complexity, the significant performance gains outweigh this drawback. The OMNIVORE encoder, pre-trained on multiple modalities, proves particularly effective for both infrared and depth modalities, outperforming even the modality-specific MAE encoders. In conclusion, the results provide compelling evidence for leveraging modality-specific pre-trained encoders in action recognition tasks. This approach achieves faster convergence and higher downstream performance, justifying the added complexity. The success of universal encoders like OMNIVORE in learning shared repre-

sentations across diverse modalities offers a promising direction for future research in multi-modal alignment and action recognition.

6.1.4 SELF-ALIGNMENT

Building upon the successful application of contrastive learning and modality-specific pre-training in our multi-modal action recognition framework, we now explore a novel approach called self-alignment. This technique aims to leverage the complementary strengths of different pre-trained models within a single modality, potentially reducing our dependence on annotated data and further enhancing the quality and generalizability of the learned representations.

Table 6.7: Retrieval Evaluation of RGB Self Alignment with Frozen CLIP-ViP (RGB) and OMNIVORE (RGB2)

	Daa(avg)		NTU(CV)		NTU(CS)	
	RGB	RGB2	RGB	RGB2	RGB	RGB2
CLIP-ViP-Base	5.78	21.32	8.39	12.11	13.6	17.14
CLIP-ViP-Fine-Tuned	56.76	54.01	71.7	72.52	74.12	74.91

Table 6.8: LEP of RGB Self Alignment with Frozen CLIP-ViP (RGB) and OMNIVORE (RGB2)

	Daa(avg)		NTU(CV)		NTU(CS)	
	RGB	RGB2	RGB	RGB2	RGB	RGB2
CLIP-ViP-Base	46.65	55.34	56.43	60.46	53.72	55.51
CLIP-ViP-Fine-Tuned	63.1	65.94	77.73	82.19	76.11	79.78

The experimental results presented in Tables 6.7 and 6.8 demonstrate the impact of self-alignment on the quality and effectiveness of RGB representations using CLIP-ViP and OMNIVORE models. In both RT and LEP evaluations, the self-aligned OMNIVORE RGB2 encoder consistently outperforms the frozen CLIP-ViP RGB encoder across all datasets, for both base and fine-tuned versions. In the retrieval evaluation (Table 6.7), the performance gap between RGB2 and RGB is more pronounced for the base version compared to the fine-tuned version. For the base version on the DAA dataset, RGB2 achieves an impressive 21.32% accuracy compared to just 5.78% for RGB, representing a significant improvement of 15.54 percentage points. Similarly, on NTU(CV) and NTU(CS), RGB2 reaches 12.11% and 17.14%, respectively, while RGB remains at 8.39% and 13.6%, indicating gains of 3.72 and 3.54 percentage points. In contrast, for the fine-tuned version, the performance difference between RGB2 and RGB is smaller but still noticeable. On the DAA dataset, RGB2

achieves 54.01% compared to 56.76% for RGB, a slight decrease of 2.75 percentage points. However, for NTU(CV) and NTU(CS), RGB2 slightly outperforms RGB with accuracies of 72.52% and 74.91% compared to 71.7% and 74.12%, representing improvements of 0.82 and 0.79 percentage points, respectively. The LEP results (Table 6.8) show a similar pattern. For the base version, RGB2 significantly outperforms RGB across all datasets, with accuracy improvements ranging from 1.79 to 8.69 percentage points. In the fine-tuned setting, RGB2 maintains its lead over RGB, even though with smaller margins of 2.84 to 4.46 percentage points. These results suggest that self-alignment provides a more substantial benefit when the CLIP-ViP RGB encoder is not fine-tuned. The fine-tuned CLIP-ViP encoder, having already captured rich, dataset-specific representations, leaves less room for improvement through self-alignment. Nonetheless, the consistent outperformance of RGB2 over RGB, even in the fine-tuned setting, highlights the potential of self-alignment to further refine and enhance the learned representations. To better understand the factors contributing to the improvements gained through self-alignment, we conduct a series of ablation experiments on the DAA split 0 dataset. These experiments aim to shed light on the role of the OMNIVORE weights, the impact of the loss function, and the properties of the resulting embedding space.

ABLATION EXPERIMENTS

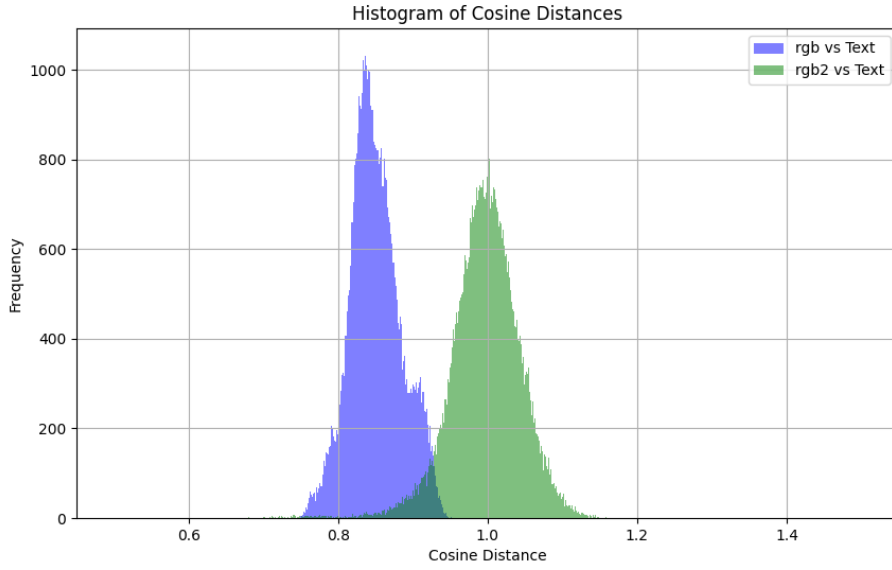
The ablation experiments in Table 6.9 provide valuable insights into the factors influencing the quality of the self-aligned representations, focusing on the analysis of the DAA split 0.

Table 6.9: Ablation Experiments on Self Alignment CLIP-ViP-Base (RGB) aligned to OMNIVORE (RGB2) on DAA split 0

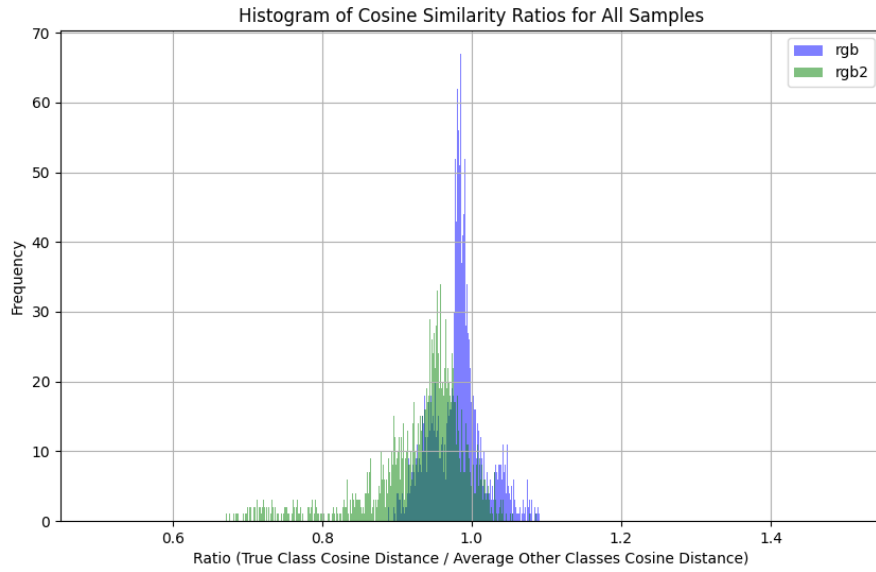
	RGB	RGB2
<i>Retrieval</i>		
OMNIVORE pre-trained	4.3	24.67
OMNIVORE scratch	4.3	10.39
ONIVORE scratch mse loss	4.3	4.05
<i>kNN</i>		
OMNIVORE pre-trained	32.55	25.55

When the OMNIVORE model is initialized from scratch (OMNIVORE scratch), the retrieval accuracy of the RGB2 embeddings drops to 10.39% compared to 24.67% when using pre-trained weights. This significant decrease in performance highlights the importance of starting with high-quality, pre-trained representations for the self-alignment task. The pre-trained OMNIVORE weights seem to provide a strong foundation of relevant visual features, which the self-alignment process can then effectively leverage, adapt, and refine to align

Figure 6.2: Compactness of the embedding space in RGB to RGB alignment for DAA split 0



(a) Distribution of cosine distances between RGB embeddings and text embeddings



(b) Distribution of cosine similarity ratios (true class cosine distance / average false class cosine distance)

with the frozen CLIP-ViP RGB embeddings. Furthermore, the comparison between the InfoNCE loss and the Mean Squared Error (MSE) loss reveals the critical role of the contrastive

learning objective in enabling effective representation learning during self-alignment. When using the MSE loss (OMNIVORE scratch mse loss), the retrieval accuracy of the RGB2 embeddings further decreases to 4.05%, only slightly lower than the RGB accuracy of 4.3%. This suggests that the MSE loss struggles to capture the rich semantic information present in the CLIP-ViP embeddings, resulting in less discriminative and less informative self-aligned representations. On the other hand, the InfoNCE loss allows the OMNIVORE RGB2 embeddings to effectively learn from the frozen CLIP-ViP RGB embeddings by maximizing the similarity between corresponding samples while minimizing the similarity between non-corresponding samples. Consequently, this corroborates the explanation that minimizing the off-diagonal values in the cosine similarity matrix does influence the benefit of self-alignment.

The distribution of cosine distances between the CLIP-ViP base RGB embeddings and the text embeddings (Figure 6.2a) is shifted to the left, indicating that these embeddings, learned on a large-scale dataset, are relatively compact and close to their corresponding text embeddings. When we align the OMNIVORE RGB2 embeddings using the InfoNCE loss and pre-trained weights, we observe a shift and flattening of the distribution, suggesting an increase in the overall cosine distances between the RGB2 embeddings and their respective text embeddings. This is in line with the kNN results presented in Table 6.9. The kNN results show that the pre-trained OMNIVORE RGB2 embeddings achieve a lower accuracy (25.55%) compared to the CLIP-ViP base RGB embeddings (32.55%), which is exactly opposite to the retrieval performance. This apparent contradiction can be explained by the fact that the kNN classifier relies on the absolute distances between embeddings, which may not capture the relative relationships between classes. The second plot (Figure 6.2b) provides a clearer picture by plotting the ratio of true class cosine distance to the average false class cosine distance. We observe that the RGB2 embeddings actually outperform the base RGB embeddings in this metric, indicating that the alignment process enhances the separability between the true class and the false classes, despite the overall increase in cosine distances. This observation is consistent with the higher retrieval performance of the RGB2 embeddings, as the retrieval task relies on the relative similarities between embeddings rather than their absolute distances.

Overall the self-alignment experiments demonstrate the effectiveness of aligning diverse pre-trained models within a single modality to improve representation quality and downstream task performance. The results confirm our hypothesis that self-alignment significantly enhances embedding quality, particularly when the base encoder is not fine-tuned. Moreover, the experiments highlight the importance of high-quality pre-trained weights for successful self-alignment and the potential of this approach to improve embedding quality without relying on traditional labels, making it valuable in label-scarce scenarios. Furthermore, the abla-

tion studies reveal the important role of the InfoNCE loss in self-alignment, as it encourages a more discriminative embedding space.

6.2 APPLYING ALIGNMENT: PRACTICAL BENEFITS AND USES

6.2.1 CROSS VIEW AND CROSS DATASET GENERALIZABILITY

The cross-dataset and cross-view evaluations in this section aim to investigate the generalizability and transferability of the aligned multi-modal embeddings obtained using our proposed methods. By aligning the additional visual modalities to the fine-tuned CLIP-ViP representations on the NTU dataset and evaluating their performance on the DAA dataset, as well as assessing the robustness of the aligned embeddings to viewpoint changes, we gain insights into the effectiveness of our alignment methods in capturing meaningful action recognition features that generalize across datasets and views.

CROSS DATASET ROBUSTNESS

The experimental results presented in Table 6.10 reveal several key findings regarding cross-dataset robustness when comparing the performance of encoders aligned on the larger, balanced NTU dataset with those aligned on the smaller, imbalanced DAA dataset. Firstly, when considering the base CLIP-ViP RGB-Encoder, the encoders aligned on the NTU60 dataset generally outperform those aligned on the DAA dataset when evaluated on the DAA dataset itself. For instance, the RGB-IR (R I) combination achieves an IR accuracy of 32.02% when aligned on NTU60, compared to 19.59% when aligned on DAA. Similar trends can be observed for other modality combinations, such as RGB-Depth (R D) and RGB-Skeleton (R S). Secondly, fine-tuning the CLIP-ViP RGB-Encoder on the NTU dataset enhances its performance on the DAA dataset compared to the base CLIP-ViP RGB-Encoder. The fine-tuned RGB-Encoder aligned on NTU60 achieves a RGB accuracy of 58.36%, significantly outperforming the base CLIP-ViP RGB-Encoder’s accuracy of 47.26%. However, it is worth noting that while this improvement is substantial, it still falls short of the RGB-Text fine-tuning approach, which achieves a higher RGB accuracy of 63.06%. Nonetheless, while the RGB-Text fine-tuning on NTU datasets transfers well to the DAA dataset, the improvement does not consistently generalize to the aligned target modalities when the alignment is performed towards the NTU fine-tuned encoder. The accuracies of the aligned target modalities (IR, Depth, Skeleton) are generally not better when aligned on NTU towards the fine-tuned RGB-Encoder compared to the alignment on NTU towards base RGB-Encoder. For example, the IR accuracy for the RGB-IR (R I) combination is 23.73% when aligned on NTU60

and fine-tuned, while it reaches 32.02% when aligned on NTU with the base RGB-Encoder. This suggests that while fine-tuning the RGB-Encoder on NTU datasets improves its performance on the DAA dataset, the transferability of the fine-tuned RGB embeddings to other modalities through alignment may be limited.

One notable difference in the results is the performance of the skeleton modality compared to other modalities. In the base setting, the gap between the skeleton accuracies for encoders aligned on DAA and NTU is relatively small compared to the gaps for IR and depth modalities. However, in the fine-tuned setting, this gap becomes more pronounced for the skeleton modality, compared to gaps for IR and depth. This suggests that the transferability may be more challenging for skeleton data due to its unique characteristics and potential variations in joint representation across datasets. Interestingly, the results also reveal that encoders aligned on NTU120 achieve only marginal improvements compared to those aligned on NTU60 when evaluated on the DAA dataset. This observation suggests that the larger size of NTU120 does not necessarily translate to significantly better transferability to the DAA dataset. The similarity in performance between NTU60-aligned and NTU120-aligned encoders indicates that the additional data in NTU120 may not provide substantial benefits for cross-dataset generalization in this specific case. In conclusion, the experimental results partially support the hypothesis that encoders aligned on the larger, balanced NTU dataset outperform those aligned on the smaller, imbalanced DAA dataset when evaluated on the DAA dataset. This is evident for the base CLIP-ViP RGB-Encoder, where the NTU60-aligned encoders achieve higher accuracies compared to the DAA-aligned encoders for various modality combinations.

CROSS-VIEW ROBUSTNESS

To assess the view-invariance of the aligned embeddings, a cross-view evaluation was performed on the DAA dataset, which provides multiple infrared views for each sample. We perform an RGB (fine-tuned CLIP-ViP) Kinect IR (OMNIVORE) alignment. A classifier was trained on the embeddings of the Kinect IR modality and then evaluated on the five additional IR views available in DAA. We compare our approach to two reference papers, I3D [MRH⁺19] and Transdarc [PRY⁺22], in terms of their robustness to view changes. The first pairing, I3D [MRH⁺19] Kinect IR and our method, involves training on the Kinect IR view and evaluating on the other views. The second pairing, I3D [MRH⁺19] Inner Mirror and Transdarc [PRY⁺22] Inner Mirror, involves training on the Inner Mirror view and evaluating on the other views. As shown in Table 6.11, our method achieves an average balanced accuracy of 20.47% when trained on Kinect IR and evaluated on the other views (excluding Kinect IR itself). This is significantly higher than I3D [MRH⁺19] trained on Kinect IR,

Table 6.10: LEP on DAA. Comparison between encoders aligned on DAA itself against encoders aligned on NTU60(CV) and NTU120(CV)

Section	Aligned Modalities	Accuracies			
		R	I	D	S
Base CLIP-VIP RGB					
Aligned on DAA (avg)	R I		19.59	-	-
Aligned on NTU60	R I		32.02	-	-
Aligned on NTU120	R I		28.57	-	-
Aligned on DAA (avg)	R D		-	15.89	-
Aligned on NTU60	R D		-	18.98	-
Aligned on NTU120	R D		-	23.96	-
Aligned on DAA (avg)	R S	47.26	-	-	31.55
Aligned on NTU60	R S		-	-	32.38
Aligned on NTU120	R S		-	-	32.56
Aligned on DAA (avg)	R I D S		23.27	16.44	28.4
Aligned on NTU60	R I D S		27.64	24.26	30.92
Aligned on NTU120	R I D S		27.48	24.73	33.98
Fine tuned CLIP-VIP RGB					
Ft + aligned on DAA (avg)	R I	63.06	23.11	-	-
Ft + aligned on NTU60	R I	58.36	23.73	-	-
Ft + aligned on NTU120	R I	58.75	28.56	-	-
Ft + aligned on DAA (avg)	R D	63.06	-	19.41	-
Ft + aligned on NTU60	R D	58.36	-	20.68	-
Ft + aligned on NTU120	R D	58.75	-	18.88	-
Ft + aligned on DAA (avg)	R S	63.06	-	-	42.03
Ft + aligned on NTU60	R S	58.36	-	-	33.10
Ft + aligned on NTU120	R S	58.75	-	-	35.20
Ft + aligned on DAA (avg)	R I D S	63.06	18.53	10.48	32.85
Ft + aligned on NTU60	R I D S	58.49	20.67	20.39	34.49
Ft + aligned on NTU120	R I D S	59.8	15.83	18.61	30.74

which attains only 9.42% accuracy on average for the other views. The 57.66% and 72.90% figures for our method and I3D respectively are their performances when evaluated on the Kinect IR view, which is the view they were trained on. This shows that I3D performs better on its trained view, but our method generalizes better to other views. This comparison demonstrates the superior robustness of our aligned embeddings to viewpoint changes compared to the I3D method, even though I3D performs better on the specific view it was trained on. When comparing Transdarc [PRY⁺22] Inner Mirror to I3D [MRH⁺19] Inner Mirror, we observe that Transdarc exhibits better performance on the view it was trained on (Inner

Mirror), achieving 93.58% accuracy compared to I3D’s 69.57%. However, both methods show significant performance drops when evaluated on other views, with averages of 13.82% for Transdarc and 6.86% for I3D (excluding the Inner Mirror view). Our method, despite being trained on a different view (Kinect IR), achieves comparable or even better performance on the other views compared to Transdarc. For instance, our method achieves 19.63% on the Ceiling view compared to Transdarc’s 20.87% and outperforms Transdarc on the Driver (19.91% vs 11.50%) and Co-Driver (29.11% vs 9.11%) views. These results highlight the view-invariance and generalizability of the aligned embeddings obtained through our approach. The strong performance across multiple viewpoints, particularly in comparison to the I3D method, underscores the practical value of the learned representations for real-world action recognition tasks. In conclusion, the cross-view evaluation on the DAA dataset demonstrates that our method is more robust to view changes compared to I3D [MRH⁺19] and exhibits comparable or better performance than Transdarc [PRY⁺22] when evaluated on views different from the trained view. However, it is important to note that there is still a significant gap between the performance on the trained view and the other views for all methods, including ours. While our approach shows promising results in terms of view-invariance, further research is needed to bridge this gap and achieve even more robust performance across different viewpoints.

Table 6.11: Average Balanced Accuracy (for columns 1-3)

Trained on →	Ours Kinect IR	I3D [MRH ⁺ 19] Kinect IR	I3D [MRH ⁺ 19] Inner Mirror	Transdarc* [PRY ⁺ 22] Inner Mirror
Evaluated on ↓				
Wheel	16.38%	4.27%	7.39%	16.40%
Driver	19.91%	9.02%	9.03%	11.50%
Co Driver	29.11%	19.79%	4.15%	9.11%
Ceiling	19.63%	7.34%	6.96%	20.87%
Inner Mirror	17.32%	6.66%	69.57%	93.58%
Kinect IR	57.66%	72.90%	6.77%	11.21%

*We cannot confirm whether Transdarc numbers are top-1 accuracy or balanced accuracy.

SUMMARY AND DISCUSSION OF THE GENERALIZABILITY RESULTS

The cross-dataset and cross-view evaluations provide insights into the generalizability of aligned multi-modal embeddings. Fine-tuning the RGB-Encoder on the NTU dataset improves its performance on DAA, though this improvement does not consistently transfer to all aligned target modalities. Interestingly, the skeleton modality shows a unique pattern in cross-

dataset performance, with smaller gaps between DAA and NTU alignments in the base setting, but more pronounced differences in the fine-tuned scenario. This suggests that skeleton data may have distinct transfer characteristics compared to other modalities, potentially due to variations in joint representations across datasets. Cross-view evaluation demonstrates our method’s robustness to view changes compared to existing approaches, although performance gaps between trained and unseen views persist.

6.2.2 RETRIEVAL AND LEP

The performance comparison between LEP and RT reveals several critical insights into the effectiveness and efficiency of the emerging alignment to the model’s language encoder. Our results (see Table 6.12) show a nuanced picture of the trade-offs between retrieval and LEP methods. Across most dataset splits, LEP consistently outperforms the retrieval method for both RGB and target modalities, though the performance gap varies. For instance, in the DAA dataset, the classifier achieves 63.40% accuracy for RGB, compared to 56.86% for retrieval. Similarly, for the Infrared modality in DAA, the classifier reaches 57.66% accuracy, while retrieval achieves 47.03%. Interestingly, in some cases, the performance gap between classifier and retrieval methods is relatively small, particularly for the RGB modality. In NTU60 (CV), the RGB modality shows a classifier accuracy of 77.68% versus a retrieval accuracy of 77.02%. This narrow gap suggests that for RGB data, the language encoder-based retrieval can indeed be competitive with the LEP, aligning with our hypothesis. However, the performance gap is often more pronounced for target modalities like infrared and depth. In NTU120 (CS), the infrared modality shows an LEP accuracy of 76.75% compared to a retrieval accuracy of 65.75%. This larger discrepancy for target modalities indicates that while the retrieval method offers efficiency and flexibility, it may come at the cost of reduced performance compared to a supervised approach, especially for the modalities that are only indirectly aligned towards the text encoder. Notably, in some dataset splits like NTU60 (CS) and NTU120 (CS), certain target modalities (infrared and depth) achieve higher accuracies than RGB in classification tasks. For instance, in NTU60 (CS), infrared reaches 80.94%, and depth achieves 80.31%, both surpassing the RGB accuracy of 76.04%. This suggests that in certain scenarios, aligned target modalities can provide more discriminative features for classification than RGB, highlighting the potential of our alignment approach. While the retrieval method generally shows lower accuracy, it offers practical advantages by eliminating the need for an additional classifier training phase. This could be particularly beneficial in scenarios where quick deployment or frequent updates are required. In conclusion, our findings partially support the hypothesis. The language encoder-based retrieval for RGB data can be competitive with LEP in some cases. However, for target modalities, the performance

gap is more significant, indicating that the LEP still holds an advantage in terms of accuracy. The choice between these methods should be based on the specific requirements of the task, balancing the need for accuracy against the benefits of quick deployment and immediate use of the language encoder without additional training. These results suggest that while the current approach shows promise, there is room for improvement in narrowing the performance gap between retrieval and classification methods, especially for target modalities.

Table 6.12: Combined Classifier and Retrieval Performance Comparison

Dataset	Target Modality	LEP Acc. (%)	Retrieval Acc. (%)	RGB LEP (%)	RGB RT (%)
DAA (avg)	Infrared	57.66	47.03	63.40	56.86
	Depth	49.29	37.99		
	Skeleton	42.03	33.61		
NTU60 (CV)	Infrared	73.36	61.09	77.68	77.02
	Depth	77.67	69.66		
	Skeleton	71.90	62.16		
NTU120 (CP)	Infrared	71.70	59.47	66.84	64.18
	Depth	67.95	58.87		
	Skeleton	59.89	51.09		
NTU60 (CS)	Infrared	80.94	71.23	76.04	68.54
	Depth	80.31	71.48		
	Skeleton	67.12	60.64		
NTU120 (CS)	Infrared	76.75	65.75	70.77	66.23
	Depth	73.01	61.78		
	Skeleton	60.10	49.60		

6.2.3 FUSION PERFORMANCE AND MISSING MODALITY SCENARIO

To assess whether our encoders capture unique, complementary features during alignment, we conducted a fully integrated alignment using the best model combinations from previous experiments. This section evaluates the performance of an attention-based fusion mechanism across multiple modalities and investigates its robustness in scenarios where certain modalities are missing, providing insights into the practical benefits of our multi-modal approach for action recognition.

The experimental results presented in Tables 6.13 and 6.14 provide valuable insights into the performance of the attention-based fusion mechanism across multiple modalities and its robustness in scenarios where certain modalities are missing.

For the NTU dataset, under both the Cross-View (CV) and Cross-Subject (CS) splits, the attention-based fusion head with all modalities present consistently outperforms the best sin-

Table 6.13: Performance Evaluation of Single Encoder and Attention Based Fusion Head (DAA). *Ft aug* denotes using our augmentations as proposed in Section 4.2.1, while also repeating the samples in minority classes to match the number of samples in the class with most samples present, during AP and LEP. *Ft+a on CV* is the cross evaluation of encoders fine tuned and aligned on NTU60 (CV)

Modality	Base	Ft	DAA(avg)	
			Ft aug	Ft+a on CV
Single Encoder Linear Evaluation				
RGB	47.17	63.24	67.16	58.59
IR	47.96	52.60	62.66	46.93
DEPTH	43.12	45.47	58.19	38.64
SKELETON	37.54	38.28	41.98	33.86
Attention Based Fusion Head with Missing Modalities				
ALL	46.20	58.58	62.08	56.31
Blocked Modalities				
RGB	42.64	49.96	61.23	42.22
IR	43.26	54.44	57.88	50.24
DEPTH	46.21	58.98	62.44	54.06
SKELETON	45.22	56.63	62.27	52.70
RGB+IR	39.34	45.13	56.29	35.99
RGB+DEPTH	43.48	51.45	61.37	39.92
RGB+SKELETON	39.95	49.04	61.27	38.94
IR+DEPTH	30.11	47.26	46.37	40.43
IR+SKELETON	40.13	52.98	58.3	45.54
DEPTH+SKELETON	43.52	55.6	63.3	47.65

gle encoder across all configurations (Base, Ft, Ft aug). In the CV split with fine-tuning and augmentations (Ft aug), the fusion head achieves an impressive accuracy of 91.63%, surpassing the highest single encoder performance of 85.41% by the IR encoder. Similarly, in the CS setting, the fusion model reaches a peak accuracy of 88.07%, again outperforming the best single encoder.

Under scenarios of blocked modalities, performance mostly declined; however, the fusion head maintained relatively high robustness. Notably, blocking the IR modality alone resulted in the least impact on performance, with accuracy scores of 86.96% and 84.76% in CV and CS splits, respectively, under the fine-tuned and augmented condition. In contrast, combinations where RGB was blocked with either IR or Skeleton modalities showed more substantial declines, indicating a higher dependency on the RGB modality.

Table 6.14: Performance Evaluation of Single Encoder and Attention Based Fusion Head (NTU CV and CS). *Ft aug* denotes using our augmentations as proposed in Section 4.2.1

	NTU (CV)			NTU (CS)		
	Base	Ft	Ft aug	Base	Ft	Ft aug
Single Encoder Linear Evaluation						
RGB	56.54	77.69	87.23	53.79	76.07	83.55
IR	66.71	80.3	85.41	66.11	79.87	83.53
DEPTH	59.29	78.29	81.42	66.33	78.93	81.41
SKELETON	63.08	71.48	72.62	59.5	66.58	67.2
Attention Based Fusion Head with Missing Modalities						
ALL	73.19	87.48	91.63	70.79	83.89	88.07
Blocked Modalities						
RGB	69.98	83.16	85.57	69.14	80.57	83.18
IR	70.35	86.94	91.4	69.32	83.51	86.96
DEPTH	72.97	87.12	91.85	69.9	83.22	87.13
SKELETON	67.71	84.78	89.86	67.55	83.13	87.34
RGB+IR	62.92	80.59	82.85	66.43	77.05	79.93
RGB+DEPTH	63.36	81.35	84.8	67.62	78.24	80.92
RGB+SKELETON	65.13	80.4	82.58	65.7	79.62	82.18
IR+DEPTH	59.22	82.73	87.72	59.72	80.04	81.46
IR+SKELETON	60.85	82.95	89.11	63.99	81.14	86.36
DEPTH+SKELETON	66.95	82.76	89.56	64.15	80.69	86.64

The DAA results paralleled the NTU findings in some aspects but differed in others. While the fusion head’s performance is competitive across all configurations, it does not match the best single-encoder performance. One plausible explanation for the fusion mechanism’s failure to surpass single modality classifiers on the DAA dataset could be the dataset’s class imbalance. However, this does not seem to fully account for the performance gap, as the discrepancy persists even after applying class balancing techniques such as repeating samples for minority classes. In addition to that the results for the cross-evaluation of encoders aligned on NTU exhibit the same discrepancy between fusion and single encoder performance. Therefore corroborating the interpretation that dataset imbalance cannot explain the bad fusion performance entirely. This suggests that other factors, such as the dataset’s inherent complexity or the specific interplay between modalities, may be contributing to the observed behavior. Another possible explanation could be that either the additional modalities do not

offer meaningful complementary information for DAA or our proposed attention-based fusion mechanism is not able to capture any complimentary features effectively.

The hypothesis stated that an attention-based fusion mechanism would outperform any individual encoders assessed with linear classifiers, particularly in terms of robustness when modalities are missing. The results from the NTU dataset strongly support this hypothesis. The fusion head not only surpassed individual encoder performances significantly but also demonstrated considerable robustness in scenarios with blocked modalities, maintaining relatively high accuracy even when one or two modalities were unavailable. However, the DAA dataset results present a more complex picture, with the fusion head showing competitive but not superior performance compared to the best single encoders. In conclusion, while the fusion mechanism’s underperformance on the DAA dataset compared to single modality classifiers raises important questions about the impact of dataset imbalance and other factors, the increased robustness demonstrated by the fusion head in the presence of blocked modalities is a promising finding. The enhanced performance and robustness of the fusion model in the NTU dataset confirm the hypothesis that integrated models leveraging multiple modalities can offer superior recognition capabilities and resilience against modality loss. However, the mixed results from the DAA dataset highlight the importance of considering dataset-specific factors and the potential need for tailored fusion strategies to optimize performance across different settings. Further research into the specific characteristics of the DAA dataset and the development of more adaptive fusion techniques may help bridge the performance gap and unlock the full potential of multi-modal fusion for action recognition in challenging real-world settings.

6.2.4 ZERO-SHOT EVALUATION ON DRIVE&ACT

The zero-shot evaluation results on the DAA dataset, as presented in Table 6.15, demonstrate the effectiveness of our multi-modal alignment approach in generalizing to novel driver behaviors. Our CLIP-ViP RGB encoder, fine-tuned on the seen classes, achieves a balanced accuracy of $43.26\% \pm 11.02\%$, significantly outperforming all the baselines from Reiß et al. [RRHS20], including their best-performing Early Semantic Fusion method, which reaches an accuracy of $29.22\% \pm 8.01\%$. Furthermore, the self-aligned OMNIVORE RGB encoder (RGB2) exhibits even better performance, attaining a balanced accuracy of $45.94\% \pm 7.91\%$. This improvement highlights the benefit of our self-alignment approach in enhancing the generalizability of the learned representations. Notably, the aligned OMNIVORE IR encoder also surpasses the baselines from Reiß et al., achieving a balanced accuracy of $40.48\% \pm 11.06\%$. This result underscores the effectiveness of our cross-modal alignment technique in transferring knowledge from the RGB modality to the IR modality, enabling superior

Table 6.15: Zero-shot evaluation results on the DAA dataset. We report the balanced accuracy (in %) averaged over 10 distinct 14-10-10 splits as proposed by [RRHS20], along with the standard deviation.

Method	Balanced Accuracy (%)
Random Baseline	10.00
CONSE [RRHS20]	28.79 ± 6.08
DeViSE [RRHS20]	27.69 ± 8.66
DAP [RRHS20]	28.66 ± 7.15
f-WGAN [RRHS20]	25.18 ± 4.17
f-WGAN + fine-tuning [RRHS20]	27.01 ± 5.27
Early Semantic Fusion [RRHS20]	29.22 ± 8.01
Late Semantic Fusion [RRHS20]	28.45 ± 6.26
CLIP-ViP (RGB) [Ours]	43.26 ± 11.02
OMNIVORE aligned to CLIP-ViP (RGB2) [Ours]	45.94 ± 7.91
OMNIVORE aligned to CLIP-ViP (IR) [Ours]	40.48 ± 11.06

zero-shot recognition performance. It is important to acknowledge a potential limitation of our study related to the pre-training dataset used for the CLIP-ViP model. The CLIP-ViP model was pre-trained on a large-scale dataset containing approximately 23,000 action classes [XSL⁺22], which might lead to some degree of dataset contamination in our zero-shot evaluation. Ideally, the pre-training dataset should be carefully curated to exclude any classes related to the target dataset to ensure a true zero-shot setting. However, due to resource and time constraints, retraining the CLIP-ViP model from scratch on a filtered dataset was not feasible within the scope of this work. Despite this potential limitation, we argue that in real-world scenarios, achieving a completely uncontaminated zero-shot setting becomes increasingly challenging as the scale and diversity of the pre-training datasets grow. As we move towards more generalizable and scalable models, the notion of true zero-shot recognition may become less practical. Instead, our focus should be on developing approaches that can effectively leverage knowledge from large-scale pre-training to enable robust performance on novel classes or domains, even if some degree of overlap exists between the pre-training and target classes or domains. In conclusion, our zero-shot evaluation results on the DAA dataset demonstrate the superiority of our multi-modal alignment approach over existing baselines, showcasing its ability to generalize to unseen driver behaviors. While the potential impact of pre-training dataset contamination cannot be entirely ruled out, we believe that our method represents a significant step towards more generalizable and scalable zero-shot recognition in real-world applications.

6.3 BEYOND STRICT LABELS: LEVERAGING LANGUAGE FOR ALIGNMENT

6.3.1 VARIETY IN DESCRIPTIONS

Table 6.16: LEP Results for fine-tuning on different RGB-Text datasets and subsequent RGB IR alignment on DAA split 0.

Epochs	WORDS		SENTENCES		6 SENTENCES		6 WORDS	
	RGB	IR	RGB	IR	RGB	IR	RGB	IR
5	67.82	54.97	69.36	58.22	68.50	57.40	68.33	59.00
10	68.65	56.92	69.40	56.31	69.13	56.55	69.34	56.42
15	69.01	58.85	68.23	56.76	69.74	57.26	68.79	55.95

In this section, we explore the impact of language diversity on the quality of the CLIP-ViP RGB-Text fine-tuning and the subsequent alignment of target modalities, specifically the IR modality. By leveraging four variants of our CLIP-ViP dataset, each with increasing levels of textual richness, we aim to uncover the relationship between language variety during fine-tuning and the quality of the resulting RGB representations and their alignability to other modalities. The LEP results for the DAA split 0, presented in Table 6.16, highlight the impact of language diversity on the alignability of the IR encoder to the fine-tuned RGB encoder. Across all text dataset variants (Words, Sentences, 6 Sentences, and 6 Words) and training epochs (5, 10, and 15), the IR encoder’s performance consistently lags behind that of the RGB encoder, which has been fine-tuned beforehand. Focusing on the RGB encoder’s performance, we observe varying trends across the different text dataset variants. In the WORDS variant, the RGB encoder’s performance increases slightly from 67.82% at 5 epochs to 68.65% at 10 epochs but then decreases to 69.01% at 15 epochs. The SENTENCES variant shows a similar pattern, with the RGB encoder’s performance peaking at 69.40% at 10 epochs, but then dropping to 68.23% at 15 epochs. The 6 SENTENCES variant exhibits more stable RGB encoder performance, with a slight improvement from 68.50% at 5 epochs to 69.74% at 15 epochs. Finally, the 6 WORDS variant shows a minor improvement in RGB encoder performance from 68.33% at 5 epochs to 69.34% at 10 epochs, followed by a small decrease to 68.79% at 15 epochs. The IR encoder’s performance, while consistently lower than the RGB encoder, shows mixed trends across the different text dataset variants. In the WORDS variant, the IR encoder’s performance improves from 54.97% at 5 epochs to 58.85% at 15 epochs. However, in the SENTENCES, 6 SENTENCES, and 6 WORDS variants, the IR encoder’s performance fluctuates without a clear trend of improvement with increased training epochs. These findings suggest that the language diversity introduced through the

different text dataset variants has a limited impact on the alignability of the IR encoder to the fine-tuned RGB encoder. Despite the varying levels of textual richness, the IR encoder struggles to bridge the performance gap with the RGB encoder, indicating that factors beyond language diversity may play a more significant role in the alignment process. Overall, the results indicate that the RGB encoder consistently outperforms the IR encoder across all text dataset versions and epochs. The RGB encoder benefits more from the alignment process and improves with more training epochs. However, the differences between the various text dataset versions (Words, Sentences, 6 Sentences, and 6 Words) are not substantial, challenging the hypothesis that more variety in descriptions would lead to better performance. Additionally, there is no clear evidence of overfitting in either the RGB or IR encoders, as performance generally remains stable or improves with more training.

In conclusion, the LEP results for the DAA split 0 underscore the challenges of aligning the IR encoder to the fine-tuned RGB encoder, regardless of the language diversity present in the text dataset variants. The IR encoder’s performance consistently trails that of the RGB encoder, highlighting the need for further research into the factors influencing multi-modal alignment and the development of more effective alignment strategies. Our initial hypothesis that greater language variety would lead to improved RGB encoder performance and facilitate better alignment and performance of the IR encoder is not supported by the current findings. However, future work could investigate the influence of language diversity on retrieval performance, especially in zero-shot settings, to determine whether increased textual variety improves the robustness and generalization of the language encoder.

6.3.2 VISION LANGUAGE MODELS FOR LABEL GENERATION

This experiment investigates the potential of using machine-generated captions from the LLaVA model as a substitute for manual annotations in fine-tuning CLIP-ViP. Our primary goal is to assess whether these synthetic labels can enhance the performance of multi-modal action recognition systems compared to using actual dataset labels or no fine-tuning at all. Tables 6.17 and 6.18 present the results of our experiments, comparing three variations of the RGB encoder: base weights (no fine-tuning), fine-tuned on LLaVA-generated captions, and fine-tuned on real captions. We evaluate these across both RGB-IR alignment and RGB-RGB2 self-alignment setups. Overall, the retrieval accuracy achieved by fine-tuning on vision-language model-generated descriptions outperforms the base weights consistently across all datasets and alignment setups. For the RGB-IR alignment, fine-tuning on LLaVA captions leads to a significant improvement in retrieval accuracy compared to the base version, particularly on the DAA dataset, where the accuracy increases from 5.91% to 20.64% for the RGB encoder and from 17.34% to 25.59% for the IR encoder. On the NTU datasets,

the improvement is less pronounced but still notable, with the RGB encoder’s accuracy increasing from 8.39% to 13.27% on NTU(CV) and from 13.6% to 14.08% on NTU (CS). The RGB-RGB self-alignment setup exhibits even more promising results when leveraging LLaVA-generated captions. The self-aligned RGB2 encoder consistently outperforms the base version across all datasets, with accuracy improvements ranging from 15.54 percentage points on DAA(AVG) to 24.66 percentage points on NTU (CV). This suggests that the self-alignment process can effectively capitalize on the semantic information captured by the LLaVA captions, leading to more discriminative and informative representations.

Table 6.17: Comparison of LLaVa generated captions vs real captions and base weights for RGB (CLIP-ViP)to IR (OMNIVORE) alignment.

<i>LEP</i>	DAA(AVG)		NTU(CV)		NTU(CS)	
	RGB	IR	RGB	IR	RGB	IR
Random Baseline	2.94	2.94	1.67	1.67	1.67	1.67
Base CLIP-ViP	46.99	45.33	56.51	56.66	53.9	56.55
Fine-tuned on LLaVa captions	53.92	49.79	67.67	73.14	69.94	75.3
Fine-tuned on real captions	64.41	57.66	77.48	73.36	75.99	80.94
<i>Retrieval</i>						
Random Baseline	2.94	2.94	1.67	1.67	1.67	1.67
Base CLIP-ViP	5.91	17.34	8.39	12.11	13.6	17.14
Fine-tuned on LLaVa captions	20.64	25.59	13.27	21.84	14.08	22.62
Fine-tuned on real captions	56.86	47.03	77.02	61.09	68.54	71.23

Table 6.18: Comparison of Llava generated captions vs real captions and base weights for RGB (CLIP-ViP)to RGB2 (OMNIVORE) alignment.

<i>LEP</i>	DAA(AVG)		NTU(CV)		NTU(CS)	
	RGB	RGB2	RGB	RGB2	RGB	RGB2
Random Baseline	2.94	2.94	1.67	1.67	1.67	1.67
Base CLIP-ViP	46.65	55.34	56.43	60.46	53.72	55.51
Fine tuned on Llava captions	51.08	56.43	74.12	81.09	70.30	74.29
Fine tuned on real captions	63.10	65.94	77.73	82.19	76.11	79.78
<i>Retrieval</i>						
Random Baseline	2.94	2.94	1.67	1.67	1.67	1.67
Base CLIP-ViP	5.78	21.32	8.39	12.11	13.60	17.14
Fine tuned on Llava captions	20.64	27.62	13.27	23.03	14.08	26.31
Fine tuned on real captions	56.76	54.01	71.70	72.52	74.12	74.91

In terms of LEP evaluation, fine-tuning on LLaVA captions achieves performance levels between the base version and the encoders fine-tuned on real captions. While the LLaVA-

trained encoders consistently outperform the base version, there is still a notable gap compared to the performance of those trained on real annotations. This suggests that using synthetic labels allows for learning useful representations, but the quality of these representations is not yet on par with those learned from manual annotations. These results partially support our hypothesis, confirming that training and aligning the encoders using LLaVA-generated captions can outperform the base version in retrieval tasks. However, the hypothesis that LLaVA-trained encoders would achieve comparable performance to those fine-tuned on real annotations in LEP evaluations is not supported by the current results, as there is still a significant performance gap. The discrepancy between the retrieval and LEP results highlights the need for further research to improve the quality of the learned representations when using synthetic labels. While the generated captions provide valuable semantic information for retrieval tasks, the learned features may not be as discriminative or generalizable as those obtained from real annotations, limiting their effectiveness in downstream classification tasks. The relatively smaller improvement in retrieval accuracy on the NTU datasets compared to DAA could be attributed to the nature of the datasets and the specific characteristics of the generated captions. The NTU datasets may contain more complex and varied actions that are harder to capture using single-frame captions, limiting the effectiveness of the current label generation approach. Additionally, the generated captions might focus more on object-level information rather than high-level action semantics, which could be more critical for accurate retrieval on the NTU datasets.

In conclusion, the experimental results demonstrate the potential of using machine-generated captions from vision-language models like LLaVA as a substitute for manual annotations in multi-modal alignment tasks. Despite the simplicity of the current label generation approach, which relies on single frames, the results show promising improvements in retrieval accuracy and competitive performance in LEP evaluations. These findings highlight the opportunity for future research to explore more sophisticated caption generation techniques, such as leveraging temporal information from video sequences, to further enhance the effectiveness of this approach.

7 LIMITATIONS

While the proposed multi-modal co-learning framework presents some interesting ideas, it suffers from several limitations that impact its generalizability, novelty, and practical applicability.

NARROW SCOPE AND LACK OF GENERALIZABILITY

One significant limitation of this thesis is its narrow scope and lack of generalizability. The experiments are conducted on only two datasets, NTU RGB+D and Drive&Act, which represent specific and controlled environments. The NTU dataset, while large, is captured in a constrained lab setting with a limited number of subjects and action classes. The Drive&Act dataset focuses solely on driver actions within a vehicle, which is a niche application area. This narrow focus limits the generalizability of the proposed approach to real-world scenarios, where action recognition systems need to handle a wide range of environments, subjects, and action categories. Without extensive evaluation on diverse and challenging datasets, it is difficult to assess the true effectiveness and scalability of the proposed framework.

LIMITED BENCHMARK DATA

Another limitation of this thesis is the lack of comprehensive comparisons with state-of-the-art methods, particularly for the Drive&Act dataset. While the experiments compare the proposed approach with variants of the same framework using different pre-trained models and alignment strategies, there is a scarcity of benchmarks for self-supervised learning on the Drive&Act dataset across all modalities. This scarcity of benchmark data makes it challenging to assess the true quality and novelty of the proposed contributions in the context of the Drive&Act dataset. Without a thorough comparison against existing self-supervised approaches, it is difficult to determine whether the reported improvements are significant or merely incremental. Furthermore, the fusion experiments and the analysis of missing modality scenarios are limited by the lack of prior work and established baselines on the Drive&Act dataset. The absence of comprehensive benchmarks for fusion performance and robustness to missing modalities on this dataset hinders the ability to draw definitive conclusions about

the effectiveness of the proposed approach in these aspects. However, it is important to acknowledge that the Drive&Act dataset was chosen because it shares the same four modalities (RGB, depth, infrared, and skeleton) captured by a Kinect sensor with the NTU RGB+D dataset. This commonality allows for direct cross-dataset evaluation and analysis of the generalizability of the proposed framework. A table comparing the achieved accuracies with some state-of-the-art methods is presented in Appendix 4, but it does not provide an in-depth analysis of each method or explain the potential limitations of these comparisons, as that would extend beyond the scope of this thesis.

INTENTIONAL USE OF PRE-TRAINED MODELS AND INCREMENTAL NOVELTY

The proposed framework relies heavily on pre-trained models, such as CLIP-ViP and OMNIVORE, for feature extraction and alignment. This reliance on pre-trained models may be seen as a limitation because good enough pre-trained models might not be available for every domain or modality. However, it is important to acknowledge that it is an intentional design choice to reduce training times and leverage the powerful representations learned by these models. The thesis's claim on novelty is corroborated by its focus on advancing existing approaches in several key areas. Firstly, the framework extends the concept of multi-modal embedding spaces to temporal data, enabling the alignment of representations from video clips and sequences. Secondly, it identifies suitable pre-trained models, such as OMNIVORE, that can effectively handle multiple modalities and provide a strong foundation for the alignment process. This identification of suitable models contributes to the body of knowledge and helps guide future research and applications in similar problem domains. Thirdly, it applies the proposed framework to the task of action recognition, an area where extensive exploration of multi-modal co-learning approaches is still limited. While the individual components of the framework are not entirely new, combining and applying these techniques to action recognition datasets represents an incremental advancement in the field.

ANALYSIS AND INTERPRETATION OF RESULTS: BALANCING BREADTH AND DEPTH

The thesis relies heavily on proxy metrics such as Linear Evaluation Protocol (LEP) accuracy and retrieval performance to assess the quality of the learned embeddings. While these metrics provide valuable insights into the effectiveness of the proposed framework, they cannot fully capture the nuances and complexities of the learned representations. It is important to acknowledge that judging the quality of embeddings is a challenging task, and this work relies on established proxy measures to gauge the performance of the proposed approach. However, the depth of analysis varies across different aspects of the thesis. In some sections, such as

the self-alignment experiments, this work provides a more in-depth analysis, exploring the impact of different factors like the quality of pre-trained weights and the choice of loss function on the learned representations. The ablation studies and visualization of embedding spaces offer valuable insights into the properties and behavior of the self-aligned embeddings. On the other hand, the analysis of fusion performance and the investigation of missing modality scenarios may be more superficial. While this work does explore the impact of dataset imbalance on fusion performance in the Drive&Act dataset and makes suggestions for further investigation, it does not provide a conclusive explanation for the observed discrepancies. It is important to recognize that this study covers a broad range of design choices and experiments, incorporating five modalities (including language) across two datasets with multiple splits and diverse pre-trained models. This breadth of exploration allows it to offer a comprehensive perspective on the potential of multi-modal co-learning for action recognition. By capturing a wide range of design choices and experimental setups, the thesis aims to gauge the potential of the proposed framework and identify promising directions for future research. While the current results may not provide definitive answers to all the complex issues explored, they contribute to a better understanding of the strengths and limitations of multi-modal co-learning approaches in action recognition. The thesis helps identify areas where more detailed analysis and focused investigations are needed, such as the impact of dataset characteristics on fusion performance and the generalizability of the learned representations across different domains and tasks.

8 CONCLUSION

This thesis introduced the Cross-Modal Video Representation Alignment (CMVRA) framework, a novel multi-modal co-learning approach for human action recognition. CMVRA leverages a video extension of the CLIP model as an anchor for aligning representations from diverse visual modalities. The experimental results demonstrate the effectiveness of this approach, particularly when combined with pre-trained models like OMNIVORE, which exhibited strong synergies with the CLIP-ViP video extension. Key contributions of this work include the introduction of self-alignment, which aligns representations within the same modality using different pre-trained model configurations, and the integration of vision-language models for artificial label generation. Self-alignment proved highly effective, even in boosting retrieval performance for non-fine-tuned models, suggesting that leveraging the complementary strengths of different pre-trained models can lead to more robust and generalizable representations, even without relying on traditional labels. The combination of self-alignment with artificial label generation using the LLaVA vision-language model shows promise in reducing reliance on manual annotations and improving the generalizability of learned representations. The CMVRA framework also demonstrates the ability to leverage the anchor model’s alignment to text, enabling an emergent alignment to language for all visual modalities. This bridges the gap between vision and language, allowing for more intuitive and interpretable representations of human actions. Although the performance gap between the emergent text alignment and the fully supervised linear evaluation protocol (LEP) is not negligible, the framework achieves competitive results without the need for extensive manual labeling. Furthermore, the zero-shot evaluation on the Drive&Act dataset further showcases the generalizability and robustness of the proposed approach regarding novel classes, with the CLIP-ViP RGB encoder, self-aligned OMNIVORE RGB encoder, and aligned OMNIVORE infrared encoder all outperforming existing baselines from the literature, demonstrating the effectiveness of the CMVRA framework in handling novel action classes. The proposed approach exhibits some robustness to dataset transfers and viewpoint changes, suggesting its potential for real-world applications. However, the performance still drops significantly when evaluating on different views and datasets, indicating that the learned representations may not be as generalizable as desired. Furthermore, when evaluated on the narrow

scope of the datasets used in this study, the CMVRA framework falls short of state-of-the-art fully supervised methods in terms of overall performance. This highlights the need for further optimization and advancements to bridge the gap between the proposed approach and the best-performing supervised techniques. While the current study does not definitively establish the real-world applicability of CMVRA in action recognition, the framework’s ability to reduce dependence on labels suggests its potential for practical use in the future, provided that the performance can be further improved. Fusion experiments on the NTU dataset showcase the benefits of integrating information from multiple modalities, with a simple attention-based fusion head yielding a significant performance boost and increased robustness to missing modalities. However, the fusion approach’s performance on the Drive&Act dataset remained lower compared to NTU, with the exact root cause of this discrepancy requiring further investigation. In conclusion, the CMVRA framework presents a promising direction for multi-modal co-learning in human action recognition. The introduction of self-alignment and the integration of vision-language models demonstrate potential for reducing the reliance on manual annotations and improving the generalizability of learned representations. Future research should focus on further optimizing the contrastive learning framework, investigating factors influencing fusion performance, and exploring the framework’s potential for practical applications in real-world scenarios.

9 FUTURE WORK

This thesis has identified four major directions for future research. As outlined in the limitations, our work is currently constrained to two datasets, which compromises the generalizability of results. Particularly, the cross-dataset evaluation is limited as it is only conducted in one direction. Future work should extend this to include more action recognition datasets and implement more sophisticated cross-evaluations. While we evaluated cross-dataset performance with CLIP-ViP exclusive encoders, our results suggest that exploring this with Omnivore is warranted. Additionally, investigating other CLIP video extensions that maintain the binding to text could yield valuable insights. The most practical application of this concept is its theoretical zero-shot ability across multiple modalities using the text encoder for retrieval. Currently, this is treated as an emergent property of alignment towards the frozen RGB encoder that has been previously aligned to text. Future research should focus on improving this alignment through modifications of the loss function or more sophisticated batch construction methods that go beyond treating all off-diagonal values as negatives. Exploring additional alignment goals, such as cross-modal reconstruction or introducing earlier layer alignment goals in the encoders, could also prove beneficial. Given the rapid progress in vision-language models, there is significant potential for improvement with current and future models. Enhancing these models could substantially increase practical applications. One promising direction is the utilization of multiple frames to enable the model to reason over the temporal dimension. Furthermore, exploring large multi-modal models capable of reasoning across tokens of multiple modalities directly could theoretically boost their ability to caption sequences of frames more effectively. Another avenue for future work is the use of synthetic data for training our framework, not only for creating artificial labels but also for generating action sequences. Regarding fusion, the failure of fusion on the DAA dataset that cannot be explained by dataset imbalance motivates further investigation. Future research should explore various fusion approaches to address the limitations observed in our current work. Careful analysis of the factors influencing fusion performance and a systematic evaluation of alternative approaches will be essential in advancing this aspect of the CMVRA framework.

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APPENDIX

I MASKED AUTOENCODER TRAINING AND RESULTS

For analyzing convergence improvements in the alignment phase, we aimed to create modality-specific pre-trained models. Therefore, we employed the OmniMAE architecture to create modality-specific pre-trained models for our multimodal action recognition tasks. This appendix details the implementation and training process of OmniMAE within our pipeline. To initialize our models, we used the pre-trained weights from `vit_base_mae_pretrained`¹, leveraging transfer learning to potentially speed up convergence and improve final performance. We trained separate MAE models for each modality (RGB, IR, and depth) and data split. During each training iteration, we randomly masked 80% of the input patches. The masked input was then processed through the OmniMAE encoder, and the full input was reconstructed using the OmniMAE decoder. We calculated the Mean Squared Error (MSE) between the reconstruction and the original input to serve as our loss function. For optimization, we utilized the Adam optimizer with its default beta parameters and an initial learning rate of 0.0001. To adapt the learning rate during training, we implemented a ReduceLROnPlateau scheduler, which reduced the learning rate by a factor of 0.1 if the validation loss did not improve for 5 consecutive epochs. We trained the models for 75 epochs with a batch size of 8. After each epoch, we evaluated the model on a validation set and saved a checkpoint if the validation loss improved. This process was repeated separately for each dataset (NTU and DAA) and each of their respective splits (CS and CV for NTU; splits 0, 1, and 2 for DAA). Upon completion of the training process, we evaluated the best-performing model (as determined by validation loss) on a separate test set to assess its final performance. This OmniMAE training process enabled us to create modality-specific pre-trained models tailored to our datasets. These models served as initial weights for our subsequent alignment experiments, intending to achieve faster convergence and improved performance in our multimodal action recognition tasks.

The test accuracies for the MAE encoders with LEP evaluation reported in Table 1 provide interesting insights into the performance of modality-specific pre-trained models. One no-

¹<https://github.com/facebookresearch/omnivore/tree/main/omnimae>

Table 1: Test Accuracy for MAE Encoders with LEP

Dataset Split	RGB	IR	Depth	Skeleton
DAA Split 0	46.31	42.11	35.21	-
DAA Split 1	44.2	34.94	33.54	-
DAA Split 2	37.83	35.98	33.08	-
NTU Cross View	20.04	21.08	66.75	-
NTU Cross Subject	25.9	43.97	68.96	-

table observation is the remarkably better performance of the depth modality on the NTU dataset compared to the DAA dataset. We speculate that this performance difference can be attributed to the use of masked depth input during the MAE pretraining process on the NTU dataset. By masking a significant portion of the input patches, the reconstruction task focuses primarily on the action-relevant pixels rather than the background. This selective reconstruction encourages the depth encoder to learn more discriminative features that are directly related to the human actions, potentially explaining the better downstream performance. In contrast, the DAA dataset does not utilize masked depth input, which may explain the relatively lower accuracies of the depth modality compared to the NTU dataset. The presence of background information in the unmasked depth input could potentially dilute the action-specific features learned by the depth encoder. While this observation highlights the potential benefits of using masked depth input for action recognition tasks, it is important to note that we did not analyze this aspect further due to time and resource constraints.

Nonetheless, it is worth noting that the MAE pretraining, even though not optimal, proved to be superior for the alignment process compared to initializing all modalities from RGB weights, as evidenced by the higher accuracies achieved by the MAE-pretrained encoders across different modalities and datasets, as shown in Table 6.6. In addition to that the test accuracies for MAE encoders aligned to a fine-tuned CLIP-VIP demonstrate a significant improvement compared to the MAE-only training results presented in Table 1, demonstrating that MAE pre-training and alignment can work well in combination. Future research could determine whether optimizing MAE training can boost the combination even further, exploring the impact of different masking strategies and the interplay between background and foreground information in depth-based action recognition, as well as investigating more advanced MAE implementations to further optimize the pretraining process for each modality.

2 PRETRAINED MODEL ADAPTATIONS AND TECHNICAL DETAILS

OMNIVORE

For the OMNIVORE model, we utilized the pre-trained weights of the omnivore_swinB_imageNet21k model, available in the official repository². We adapted the prediction heads for the distinct datasets by replacing them with a linear layer to match the embedding dimension of other model types. RGB and IR inputs to OMNIVORE are passed as 3-channel input, while depth is repeated across 4 channels.

DINOv2

To adapt the DINOv2 model for our alignment setup, we extract features from individual video frames using the pre-trained DINOv2 image encoder. These frame-level features are then pooled along the temporal dimension and projected with a linear layer into our shared embedding dimension. This process ensures that the features are compatible with our alignment framework.

TAHAR

For the skeleton modality, we utilized the TAHAR (Temporal Adaptive Hierarchical Autoencoder for Skeleton) model [LZM⁺24], which is specifically designed for unsupervised 3D skeleton-based action recognition. The pre-trained weights for TAHAR are available in our GitHub repository³.

To adapt the skeleton data to match the expected input shape of the TAHAR model, we preprocessed the data to reshape the input skeleton data from [B, T, 25, 1, 3] to [B, T, 75], where B is the batch size, T is the number of frames, and 25 is the number of joints. This preprocessing step ensures compatibility with the TAHAR model’s expected input format.

²<https://github.com/facebookresearch/omnivore/tree/main/omnivore>

³<https://github.com/Cappl1/CMVRA/tree/master>

3 HARDWARE REQUIREMENTS

For our experiments, we utilized multiple GPU servers:

- One server with 4× NVIDIA A6000 GPUs
- Two servers with 8× NVIDIA A100 GPUs

All our experiments can in theory be conducted on the 4× A6000 cluster, the larger clusters were utilized based on availability. The hardware requirements varied depending on the specific phase of the experiment:

ALIGNMENT PHASE

- Binary one-way alignments: 80 GB of GPU memory
- Mutual adaptability or fully integrated alignment: 160 GB of GPU memory

PAIRWISE ALIGNMENT EXAMPLE GPU HOURS

- On NTU dataset (full-size videos): Approximately 20 hours for ten epochs
- On NTU dataset (lower resolution variant): Approximately 5 hours for ten epochs
- On DAA dataset: Approximately 80 minutes for ten epochs

VISION-LANGUAGE MODEL

LLaVA-v1.6-34b model with 8-bit quantization: Approximately 40 GB of GPU memory

EVALUATION PHASE

Less than 20 GB of GPU memory

PARALLELIZATION

For our multi-GPU setup, we employed PyTorch’s `DataParallel` for parallelization. This choice was primarily motivated by its straightforward implementation and lower setup complexity compared to alternatives like `DistributedDataParallel`. While `DataParallel` met the needs of our current experimental setup, we acknowledge that transitioning to `DistributedDataParallel` in future work could potentially yield improvements in efficiency and scalability, particularly for larger-scale experiments or production environments.

All reported figures are based on processing 12 frames sized to 224×224 pixels with a batch size of 32.

4 ADDITIONAL RESULT FIGURES

6.1.2 PAIRWISE VS. FULLY INTEGRATED ALIGNMENT

Table 2: LEP Accuracies for CV120 on NTU

Frozen RGB Model	Aligned Modalities	Accuracies			
		R	I	D	S
CLIP-VIP base	R I		32.9	-	-
	R D	45.07	-	25.24	-
	R S		-	-	49.38
	R I D S		54.17	35.72	50.94
CLIP-VIP fine tuned	R I		52.37	-	-
	R D	66.87	-	39.61	-
	R S		-	-	59.89
	R I D S		11.55	34.49	53.85

Table 3: LEP Accuracies for CS120 on NTU

Frozen RGB Model	Aligned Modalities	Accuracies			
		R	I	D	S
CLIP-VIP base	R I		51.83	-	-
	R D	49.29	-	26.91	-
	R S		-	-	47.47
	R I D S		59.26	45.7	49.21
CLIP-VIP fine tuned	R I		72.55	-	-
	R D	70.65	-	54.66	-
	R S		-	-	60.1
	R I D S		44.85	45.42	54.9

6.1.3 LEVERAGING MODALITY SPECIFIC PRETRAINED ENCODERS

Table 4: LEP for different Encoder Models aligned to a frozen base CLIP-ViP

Dataset	Modality	Architecture	Test Accuracy	RGB Accuracy
DAA (Average)	Infrared	CLIP-ViP	19.6	46.89
		DINO	8.82	
		DINO (fro)	15.58	
		MAE	33.12	
		OMNIVORE		
	Depth	CLIP-ViP	16.33	
		MAE	36.37	
		OMNIVORE	38.96	
	Skeleton	CLIP-ViP	31.55	
		TAHAR	33.87	
NTU (CV)	Infrared	CLIP-ViP	54.00	47.4
		OMNIVORE	56.66	
	Depth	CLIP-ViP	25.99	
		MAE	53.68	
		OMNIVORE	54.83	
	Skeleton	CLIP-ViP	62.56	
		TAHAR	60.91	
NTU (CS)	Infrared	CLIP-ViP	53.93	54.75
		OMNIVORE	56.55	
	Depth	CLIP-ViP	31.33	
		OMNIVORE	55.01	
	Skeleton	CLIP-ViP	55.66	
		TAHAR	56.35	
NTU 120 (CP)	Infrared	CLIP-ViP	32.90	45.03
		OMNIVORE	50.87	
	Depth	CLIP-ViP	25.24	
		OMNIVORE	40.66	
	Skeleton	CLIP-ViP	49.38	
		TAHAR	46.46	
NTU 120 (CS)	Infrared	CLIP-ViP	51.83	49.47
		OMNIVORE	53.95	
	Depth	CLIP-ViP	26.91	
		OMNIVORE	50.19	
	Skeleton	CLIP-ViP	47.47	
		TAHAR	60.10	

Table 5: Comparison of SOTA and best-achieved accuracies for modalities for different dataset splits. We cannot find comprehensive up to date leaderboards for every split and modality and we might have missed certain models in our search.

Dataset	Supervision	Modality	SOTA	Best Accuracy
DAA	Fully Supervised	RGB	62.95 [MRH ⁺ 19]	-
		IR	64.98 [MRH ⁺ 19]	-
		DEPTH	60.52 [MRH ⁺ 19]	-
		SKELETON		-
	Self-Supervised	RGB		65.94
		IR		57.66
		DEPTH		49.29
		SKELETON	26.03 [LEBM ⁺ 24]	42.03
NTU (CV)	Fully Supervised	RGB	95.4 [ZWW ⁺ 22]	-
		IR		-
		DEPTH	96.2 [ZWW ⁺ 22]	-
		SKELETON	98.3 [DWCL22]	-
	Self-Supervised	RGB		87.23
		IR		85.41
		DEPTH		81.42
		SKELETON	85.4 [PBDB22]	72.62
NTU120 (CP)	Fully Supervised	RGB		-
		IR		-
		DEPTH		-
		SKELETON	89.1 [SZSW22]	-
	Self-Supervised	RGB		66.83
		IR		71.7
		DEPTH		67.95
		SKELETON	64.8 [LZM ⁺ 24]	59.89
NTU (CS)	Fully Supervised	RGB		-
		IR		-
		DEPTH	80.4 [SKR ⁺ 22]	-
		SKELETON	91.5 [SKR ⁺ 22]	-
	Self-Supervised	RGB		83.55
		IR		83.53
		DEPTH		81.41
		SKELETON	72.4 [LZM ⁺ 24]	67.2
NTU120 (CS)	Fully Supervised	RGB		-
		IR		-
		DEPTH		-
		SKELETON	94.3 [WWT ⁺ 23]	-
	Self-Supervised	RGB		70.87
		IR		76.75
		DEPTH		73.01
		SKELETON	65.1 [LZM ⁺ 24]	60.1

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Erklärung zur Abschlussarbeit
gemäß § 34, Abs. 16 der Ordnung für den Masterstudiengang
Wirtschaftsinformatik vom 01. April 2019

Hiermit erkläre ich

Rothenburger Bastian

(Nachname, Vorname)

Die vorliegende Arbeit habe ich selbstständig und ohne Benutzung anderer als der angegebenen Quellen und Hilfsmittel verfasst.

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